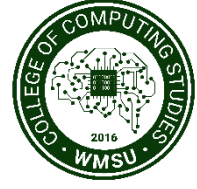




Republic of the Philippines
Western Mindanao State University
College of Computing Studies
DEPARTMENT OF INFORMATION TECHNOLOGY
Zamboanga City



Present and Ready: A Liveness Detection for Real Time Face Recognition

Attendance

A Capstone Project Presented to the Faculty of
Department of Information Technology
College of Computing Studies

In Partial Fulfillment of the Requirements for the Degree of
Bachelor of Science in Information Technology

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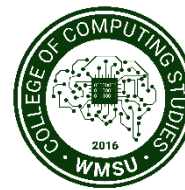
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May 2026



Republic of the Philippines
Western Mindanao State University
College of Computing Studies
DEPARTMENT OF INFORMATION TECHNOLOGY
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Approval Sheet

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74

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130 obstacle they faced.

Abstract

The study titled "Present and Ready: A Liveness Detection for Real-Time Face Recognition Attendance" was conducted to automate attendance monitoring in classroom settings. The system integrates YOLOv8 for face and body detection, InsightFace for 512-dimensional facial embedding, TorchReID for continuous identity tracking, and an ONNX-based anti-spoofing model augmented with Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) heuristics for liveness validation. Twelve domain-specific heuristic rules are layered over the deep learning pipeline to enforce operational safeguards. The system was evaluated across 12 class sessions involving 15 participants, yielding 432 face recognition attempts. Results show 85.2% face detection accuracy, 85% recognition precision, 89.8% liveness detection accuracy, and a 2.1% false acceptance rate. A one-week pilot deployment achieved an average attendance accuracy of 82.67%. A user evaluation based on ISO/IEC 25010 attributes yielded an overall satisfaction score of 4.56/5 (91.2%). The results confirm that integrating liveness detection with real-time facial recognition provides a secure, efficient, and practical solution for classroom attendance monitoring in academic institutions.

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CHAPTER I

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INTRODUCTION

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Background of the Study

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The College of Computing Studies at Western Mindanao State University still relies on traditional methods of recording attendance, and both approaches often take away valuable class time. One common method is rolling call. Instructors usually spend around 2 minutes reading names aloud and waiting for students to respond. At first, this does not seem like much, but when repeated every session, those minutes slowly eat into teaching time. This is time that could have been used to explain concepts in programming, database systems, or computer vision. Another method is the manual sign-sheet system. In this approach, students write their names one by one each taking less than 30 seconds, this may not seem much again, but it can disrupt their focus in listening to their class. A class of 30-35 students often takes 5 to 8 minutes to complete the process. Over an entire semester, these minutes turn into hours that could have been spent on deeper classroom discussions. [1].

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Paper attendance sheets often get lost, torn, or smudged. There is no centralized digital record to recover or verify lost data. When this happens, students who attended class regularly may be marked absent, leading to confusion and disputes during grading. A simple clerical error or missing sheet can unfairly impact a student's academic standing.

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Another major concern is proxy attendance. In larger classes with 30 to 35 students, some students mark their absent classmates as present. Manual methods cannot always verify these entries. Based on my four years as a college student in WMSU, this practice is common. Students often do it to help their friends avoid penalties for absences or

414 tardiness. Unfortunately, this compromises the fairness and integrity of the academic
415 process.

416 The current system also lacks the ability to monitor whether students remain present
417 throughout the session. Leaving early without permission can pose safety risks and affect
418 the quality of learning. However, the paper-based system offers no way to track early exits
419 or ensure full attendance.

420 Faculty members are equally affected by these inefficiencies. Instructors often spend an
421 extra 3 to 5 minutes after class checking attendance records or addressing student
422 complaints. This is time that could be used for preparing lessons, conducting research, or
423 mentoring students. The manual system puts unnecessary strain on teachers who should
424 be focused on delivering quality education.

425 It is ironic that CCS, a college that teaches cutting-edge technologies such as machine
426 learning and computer vision, still relies on outdated administrative methods. This project
427 proposes the development of an automated attendance system using the vision of col DS-
428 2CD2086G2-I(U) 4K AcuSense Fixed Mini Bullet Network Camera [2]. The system will
429 use real-time face recognition and liveness detection to identify students accurately while
430 preventing proxy attendance [3].

431 The attendance data will be stored in a cloud-based MySQL database [4]. A web interface
432 [5] will allow faculty and administrators to access reports and monitor attendance in real
433 time. This solution supports WMSU's digital transformation roadmap and aligns with
434 CCS's goal of applying practical IT tools in real-world scenarios. By automating attendance
435 [6], the project will improve efficiency, reduce fraud, and create a more accurate and
436 secure tracking system [7].

437 **Statement of the Problem**

438 The paper-based attendance system currently in use at the College of Computing Studies
439 (CCS) of Western Mindanao State University (WMSU) has become a source of various
440 academic and administrative difficulties, which negatively affect the overall quality of
441 education.

442 One of the biggest concerns in class sessions is the time wasted on manual attendance.
443 Instructors usually spend around 2 minutes on a quick roll call or 5 to 8 minutes passing
444 around paper sign-in sheets. This might not seem like much time, but it adds up quickly
445 when repeated every meeting throughout a semester. By the end, the loss can reach
446 between 1 to 5% of total class hours. The impact is even more noticeable in classes that
447 focus on complex and hands-on topics such as algorithms, software development, or
448 system design. These subjects require sustained attention, and losing valuable minutes
449 at the start of every meeting can break focus and reduce productivity. Students may find
450 themselves rushing through important exercises because the class started late, while
451 instructors struggle to cover all the planned material within the remaining time. Paper
452 attendance also brings other problems. Sheets can be misplaced, damaged, or even left
453 unsigned by students who forget to pass them along. Imagine a student who attended
454 class faithfully but ends up marked as absent simply because the paper did not reach
455 them. Without a centralized or digital backup, small errors like this create bigger issues
456 later, especially during grading or official record checks. These challenges show why a
457 more efficient attendance system is needed. By replacing outdated manual methods with
458 technology-driven solutions, classes can start smoothly, maximize learning time, and
459 minimize errors. This ensures that both students and instructors make the most of every
460 meeting.

461 Another issue lies in the way records are managed. Paper-based attendance sheets are
462 often lost, incomplete, or damaged. Since there is no digital backup system in place,
463 recovering accurate attendance data becomes difficult. This is especially problematic in
464 courses where attendance plays a significant role in final grade calculations. The lack of
465 reliable records can lead to disputes and unfair grading outcomes.

466 The system also raises concerns about academic integrity. In large classes of about 30 to
467 35 students, proxy attendance is common practice. Without a way to verify who is present,
468 students are often able to sign in for their absent classmates. This undermines fairness
469 and reduces the credibility of academic assessments.

470 Safety and accountability are further compromised by the system's inability to track early
471 departures. There is no mechanism to monitor or record these early exits, which creates
472 gaps in accountability and raises safety risks.

473 Faculty members are also burdened by the current process. After each class, instructors
474 often spend an additional ten to fifteen minutes verifying attendance or addressing student
475 complaints. This administrative load takes away from time that could be better spent
476 preparing lessons, conducting research, or mentoring students.

477 To address these challenges, this study aims to explore possible improvements with
478 modern technology. Specifically, it will examine the following questions:

- 479 • Can face recognition and liveness detection features help prevent proxy
480 attendance in large classes with varying session lengths?
- 481 • How much instructional time can be recovered by using an automated system?

482 This research proposes the development and implementation of an online attendance
483 system that uses real-time face recognition with liveness detection. The system will be
484 visioned by the Hikvision DS-2CD2086G2-I(U) 4K AcuSense Fixed Mini Bullet Network
485 Camera and supported by a cloud-based MySQL database. The solution aims to improve
486 accuracy, reduce fraud, lessen faculty workload, and provide video archives for auditing
487 purposes. It is designed to function effectively across classes of different durations,
488 whether 1-hour lectures [8].

489 **Objectives**

490 **General Objectives**

491 To design and develop a real-time face recognition attendance system with liveness
492 detection capable of achieving at least 85% face recognition accuracy, 90% spoof
493 detection accuracy, and ≤ 10.35 seconds attendance logging latency.

494 **Specific Objectives**

- 495 1. To integrate YOLOv8 for face detection, InsightFace for recognition, and an
496 ONNX-based model for liveness detection in a unified system.
- 497 2. To design a system architecture that could support one classroom environment
498 (10–15 students) using a 4K IP camera with RTSP streaming.
- 499 3. To evaluate the performance of the system based on accuracy (detection,
500 recognition, liveness), speed (FPS, latency), and consistency under varied
501 lighting, seating, and occlusion conditions.
- 502 4. To implement a dashboard that facilitates real-time monitoring, attendance
503 review, and automated report generation for faculty.

504 5. To conduct a user satisfaction assessment based on ISO/IEC 25010 quality
505 attributes, targeting a minimum satisfaction rating of **80%**.

506 **Scope and Limitations**

507 **Scope**

- 508 • **Environment:** One CCS classroom measuring 60–70 square meters
- 509 • **Population:** 10-15 concurrent students
- 510 • **Camera:** Hikvision DS-2CD2086G2-I(U), 4K, 20 FPS, RTSP stream
- 511 • **Distance Range:** 1–3 meters effective recognition distance
- 512 • **Performance Targets:**
 - 513 ○ ≥85% face detection & recognition accuracy
 - 514 ○ ≥90% spoof/liveness detection accuracy
 - 515 ○ ≥20 FPS real-time processing
 - 516 ○ ≤10.35 seconds logging latency
- 517 • **Attendance Categories:** Present, Late, Missing, Excuse, Absent
- 518 • **Deployment Duration:** One-week classroom pilot testing
- 519 • **Outputs:** Attendance logs, timestamps, captured frames, and faculty dashboard
- 520 analytics

521 **Limitations**

- 522 • Large lecture halls exceeding 70 sqm
- 523 • More than 15 concurrent students
- 524 • Poor lighting conditions such as backlighting, shadows, or dim illumination
- 525 • Recognition of faces covered with masks, helmets, or large accessories
- 526 • Operation beyond the 1–3 meters optimal distance
- 527 • Offline usage (requires a stable 5–15 Mbps connection)
- 528 • Multi-room or campus-wide tracking
- 529 • Detection of advanced 3D masks or deepfake spoofing
- 530 • Cameras below 4K resolution
- 531 • Long-term database scaling beyond the pilot period
- 532 • Hardware budget exceeding 10,000 pesos.

533 **Significance of the Study**

534 Previous studies and existing systems primarily concentrated on face recognition for
535 attendance but lacked robust liveness detection mechanisms. This gap leaves such
536 systems vulnerable to spoofing attacks involving printed photos, videos, or digital displays.
537 This study introduces a real-time liveness detection module that ensures the authenticity
538 of the person being recognized, thereby strengthening identity verification.

539 For CCS, the system modernizes classroom operations by reducing attendance time,
540 preventing proxy signing, and improving the accuracy of academic records. Faculty
541 members benefit from automated logging and centralize digital recordkeeping, while
542 students gain a fair and secure attendance process. On an institutional level, the system
543 supports digital transformation initiatives by providing reliable analytics and automated
544 reporting.

545 Definition of Terms

546 Table 1: Definition of Terms

Term	Definition
1. Liveness Detection	A security feature that checks if the detected face belongs to a live person and not from a photo, video, or mask.
2. Anti-Spoofing	A protective measure used to prevent fake attendance attempts through printed photos, videos, or digital displays.
3. YOLOv8	An artificial intelligence model used in the system to detect faces in real time, outputting bounding boxes and accuracy scores.
4. OpenCV	A computer vision library integrated into the project to handle image processing and assist in detecting faces.
5. Dlib	A tool that supports face tracking in the system by maintaining consistent recognition of students through persistent IDs.
6. SORT Algorithm	The tracking method used to assign IDs to faces so that students can be recognized continuously across video frames.
7. ByteTrack	A high-performance multi-object tracking algorithm used to maintain accurate tracking of multiple detected faces across frames. It improves stability and accuracy even when several students appear in the camera's view.
8. InsightFace	A deep learning model that produces a 512-dimensional facial embedding to uniquely represent each student's identity.
9. ONNX (model_rgb.onnx)	A security layer that checks whether the detected face comes from a real, live person rather than a spoof attempt (like a printed photo, phone screen, or video).
10. Facial Embedding	A unique numerical representation of a student's face that allows the system to distinguish one person from another.
11. RTSP Streaming	A network protocol that connects the Hikvision camera to the system so it can transmit live classroom video.

Term	Definition
12. MySQL Database	A cloud-based storage system where attendance records, timestamps, and student facial data are kept.
13. Flask	A lightweight web framework used to develop the web interface for managing attendance data.
14. Attendance Monitoring System	The automated process in this study that records students as Present, Late, Missing, Excuse or Absent during class sessions.
15. Biometric Data	The facial features collected from students and stored in the system for identification and attendance tracking.

547

548

CHAPTER II

549

REVIEW OF RELATED LITERATURE

550 Introduction

551 This chapter presents a comprehensive review of literature and related studies on
552 attendance monitoring systems, face recognition technologies, liveness detection, and
553 recent advancements in artificial intelligence applied to education. The discussion is
554 organized thematically to highlight how prior works inform the development of this study,
555 *“Present and Ready: A Liveness Detection for Real-Time Face Recognition Attendance.”*

556 The purpose of this review is to identify the trends, theoretical foundations, and
557 technological advancements in automated attendance monitoring. It also aims to
558 determine existing research gaps, particularly the limited implementation of **liveness**
559 **detection mechanisms** in classroom-based recognition systems and to justify the
560 development of a system that enhances both reliability and authenticity of attendance
561 records.

562 Attendance Systems in Educational Contexts

563 Attendance monitoring remains a vital component of classroom management. Traditional
564 methods such as roll calls and sign-in sheets have been widely criticized for inefficiency
565 and error proneness [9]. Digital innovations have emerged to address these concerns. For
566 example, Reynoso and Torres developed Tempus, a cloud-based attendance system in
567 the Philippines that significantly reduced roll call time while improving record accessibility
568 [10]. However, Tempus did not include advanced anti-fraud measures leaving it vulnerable
569 to proxy attendance.

570 Local implementations such as the Pilar National High School project applied facial
571 recognition for automated logging [11]. While these systems reduce manual effort, they

572 generally lack liveness detection and are therefore susceptible to photo and video
573 spoofing attacks. This gap highlights the need to integrate anti-spoofing techniques into
574 classroom-ready attendance systems.

575 Several recent system-level studies propose practical attendance architectures relevant
576 to classroom deployment. AttenFace introduced a snapshot-based approach that uses
577 periodic classroom images to determine presence throughout the session, reducing
578 computational load while improving accuracy [12]. Class-level proposals and embedded
579 implementations demonstrate how attendance engines can be decoupled from back-end
580 record keeping and integrated into institutional dashboards [13], [14]. Additionally, a cloud-
581 assisted pipeline for large venues adapted to variable lighting and crowd density,
582 suggesting scalability potential for academic spaces [15].

583 These applied systems demonstrate feasibility and administrative benefit, but they rarely
584 include robust presentation attack detection or studies under spoofing conditions. Thus,
585 while they improve convenience, they still leave academic integrity exposed a critical
586 challenge the present study aims to address.

587 **Face Recognition Technologies**

588 Face recognition has rapidly evolved with deep learning and large-scale benchmark
589 datasets. The Masked Face Recognition challenge and associated InsightFace track
590 documented significant performance drops when faces are occluded by masks, prompting
591 dataset expansions and model adaptations [16]. The CelebA *Spoof* dataset provided
592 large-scale annotated images for training anti-spoofing detectors and encouraged the
593 community to address presentation attacks using data-driven methods [17].

594

595 Practical recognition toolkits and detection engines underpin many attendance projects.
596 InsightFace provides pre-trained embedding models and deployment utilities widely used
597 for applied recognition systems [18]. Real-time object detectors such as YOLOv8 have
598 improved detection speed and made real-time face recognition feasible on lightweight
599 servers or edge devices [19]. Transformer-based models like FM-ViT offer modal flexibility
600 and improved fusion strategies for multi-input recognition [20].

601 Despite these advances, many classroom deployments use simplified recognition
602 pipelines optimized for accuracy under controlled conditions, often ignoring realistic
603 occlusion or spoofing scenarios. This misalignment underscores the need for hybrid
604 systems that integrate both recognition and liveness detection modules, as proposed in
605 the present study.

606 **Liveness Detection and Anti-Spoofing Approaches**

607 Liveness detection also called presentation attack detection is essential to prevent proxy
608 attendance. Several recent surveys and technical reports summarize the field and its
609 evolution. Yu et al. provided a comprehensive survey of deep learning-based face anti
610 spoofing summarizing texture motion and multimodal methods and pointing out
611 generalization issues in real world deployment [9]. Additional studies on deepfake and
612 forgery detection emphasized the need for physiological and temporal cues to distinguish
613 genuine from spoofed presentations [21].

614 Methodologically, researchers have combined spatial-temporal cues and depth inference
615 to improve robustness. Wang et al. proposed deep spatial gradient and temporal depth
616 learning methods that fuse per-frame features with pseudo-depth supervision to detect
617 print and replay attacks effectively [22]. Building on these works, Wang et al. [23]
618 introduced a consistency regularization framework for deep face anti-spoofing that

enhances generalization by enforcing feature stability across diverse spoof types and acquisition conditions. This approach demonstrated improved robustness under cross-dataset evaluations, addressing one of the major weaknesses of traditional supervised training in presentation attack detection. FM-ViT and multimodal contrastive learning approaches demonstrated that transformer and contrastive frameworks can fuse modalities and tolerate missing inputs while improving cross-dataset performance [20]. RGB-D based models trained with domain adversarial components also improved generalization to diverse lighting and capture conditions [24].

Physiological cues such as remote photoplethysmography (rPPG) have been explored to detect subtle blood flow variations present in live faces but absent in spoof presentations [25], [26], [27]. Industry systems such as Intel's FakeCatcher demonstrated the practical potential of blood-flow analysis for real-time deepfake detection [28].

Although these advances are promising, most liveness detection methods are validated on benchmark datasets and not evaluated in real classroom settings. Factors such as varied lighting, camera placement, and computation constraints remain underexplored, an essential motivation for integrating and testing such techniques in the educational context.

Cloud and Web-Based Integration

Cloud based back ends provide centralized storage for attendance logs and support reporting and analytics. Prior local systems including Tempus and project-based deployments commonly used centralized databases for record retrieval and administrative reporting [10], [11]. Modern attendance projects show how the recognition and liveness pipelines can be decoupled from web dashboards enabling flexible deployments that push alerts and reports to faculty and administrators [13], [15].

642 There is a design choice trade-off between on device inference which reduces network
643 dependency and cloud inference which enables heavier models and centralized updates.
644 Scalability and privacy are important design constraints in university environments and
645 must be balanced with detection performance.

646 **Local and Foreign Implementations**

647 Foreign research primarily contributes technical advancements in face recognition,
648 liveness detection, and anti-spoofing algorithms, while local studies demonstrate the real-
649 world constraints of implementation in educational institutions. International works such
650 as those by Yu et al. [9], Zhang et al. [17], Wang et al. [22], and Kim and Kwon [24]
651 establish the theoretical and methodological foundation for robust face recognition and
652 presentation attack detection. These studies develop models that achieve high accuracy
653 across benchmark datasets and explore multimodal or depth-based learning strategies
654 that enhance generalization across different environments.

655 In contrast, local research efforts such as the Tempus attendance system by Reynoso
656 and Torres [10] and the Pilar National High School prototype [11] show that the Philippines
657 has already begun to adopt intelligent attendance solutions. However, these systems
658 largely focus on recognition accuracy and data management rather than liveness
659 detection or spoofing resistance. Their implementations highlight key contextual
660 challenges including inconsistent lighting, limited processing hardware, and internet
661 connectivity issues that can affect system reliability.

662 While foreign studies excel in algorithmic sophistication, they are rarely tested in real
663 classroom settings. Conversely, local systems emphasize practicality but lack the
664 advanced deep learning and anti-spoofing components that ensure authenticity of
665 attendance data. This mismatch forms a critical gap that the present study aims to

666 address. By integrating advanced liveness detection methods derived from recent global
667 research [22], [24], [15], [9], and adapting them for classroom deployment in a local
668 environment, the proposed system seeks to bridge the divide between theoretical
669 robustness and contextual usability.

670 The synthesis of these findings suggests that a hybrid approach, one that combines the
671 methodological rigor of international research with the contextual relevance of local
672 implementation, offers the most effective path toward a secure, scalable, and practical
673 real-time attendance monitoring solution.

Table 2: Systematic Synthesis

Study	Focus Method	Findings	Identified gaps
[9]	Survey of deep learning face anti spoofing	Taxonomy of texture motion and multimodal methods	No classroom application
[10]	Tempus cloud-based attendance	Reduced manual work improved reporting	No liveness detection
[11]	School face recognition attendance prototype	Automated logging reduced manual errors	No anti spoofing
[16]	Masked Face Recognition challenge	Masks degrade recognition accuracy	Not tested for attendance use
[17]	CelebA Spoof dataset	Large, annotated spoof dataset for training	Benchmarks only not classroom tested

[22]	Spatial gradient temporal depth PAD	Stronger detection on benchmark attacks	Heavy compute for real time low resource settings
[23]	Consistency regularization PAD	Improved domain robustness	Not evaluated for classroom attendance
[24]	RGB D domain adversarial PAD	Improved generalization across capture domains	Requires depth sensors
[28]	Blood flow based deepfake detection	Real time physiological detection	Proprietary and focused on deepfakes not attendance
[2]	4K AcuSense camera datasheet	High resolution capture for recognition	Commercial cost and integration limits

[12]	Snapshot based attendance from classroom images	Scales with concurrent classes reduced computation	Liveness not included in snapshots
[13]	Class attendance architecture paper	Real time pipelines and snapshot strategies	Does not test PAD methods
[14]	Raspberry Pi and web back end	Demonstrates edge device feasibility	Limited model complexity
[15]	Cloud assisted architecture	Scales to large occupancy and varied lighting	Attendance specific evaluation limited
[21]	Survey of deepfake detectors and trends	Importance of temporal and frequency cues	Not attendance focused
[25]	Appearance and rPPG joint benchmark	Shows rPPG helps joint detection	Requires quality video for rPPG

[26]	Remote photoplethysmography for spoof detection	Effective for video-based attacks	Vulnerable in low light or occluded faces
[27]	Data augmentation and pseudo negatives	Improved generalization	More training overhead
[20]	Transformer based fusion across modalities	Flexible modal operation good performance	Higher compute needs
Proposed Study	Real time attendance with liveness	Integrates recognition and reporting PAD cloud	To be validated in WMSU classrooms with masks occlusions and resource constraints

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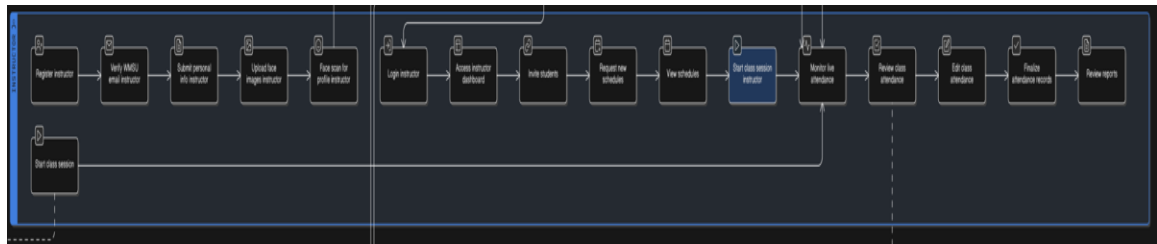
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schedules when needed, and even initiate class sessions alongside the admin. This makes the system flexible and not entirely dependent on one individual.



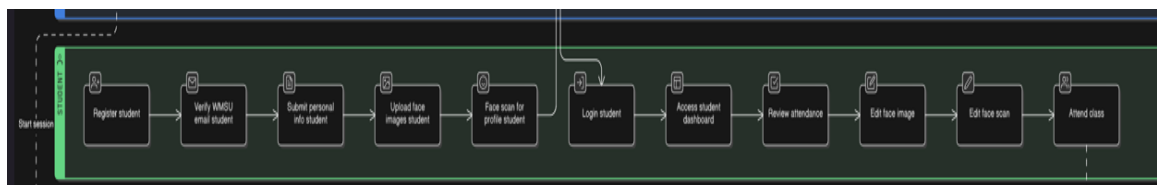
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693 Figure 3: Conceptual Framework Diagram - Instructor Registration and Setup

694 PHASE 2 – INSTRUCTOR REGISTRATION & SETUP

695 Instructors begin by registering with the system. They verify their official WMSU email,
696 provide personal details, upload face images, and complete a live face scan (includes
697 video-based data collection). Once their profile is ready, the admin reviews their
698 registration and gives final approval.

699 After being approved, instructors gain access to their dashboard. From here they can
700 invite students to join, request new schedules, and view the classes they will handle.
701 When the time comes, instructors can start class sessions, monitor attendance in real
702 time, and later review the data collected. They can edit attendance if mistakes occur,
703 finalize the records, and generate reports that reflect the actual class turnout.



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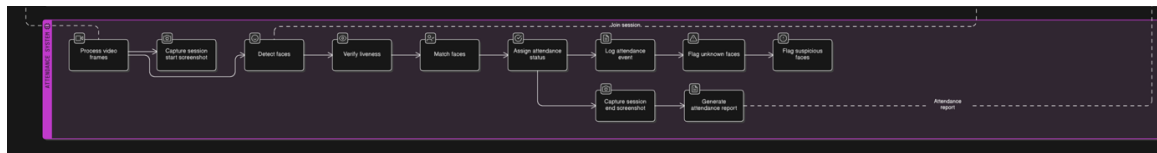
705 Figure 4: Conceptual Framework Diagram - Student Registration and Preparation

706 PHASE 3 – STUDENT REGISTRATION & PREPARATION

707 Students follow a very similar path to instructors. They register first, verify their WMSU
708 email, share personal details, upload face images, and complete their live face scan

709 (includes video-based data collection). Their registration is reviewed and approved by
710 the admin, after which they receive dashboard access.

711 Inside the dashboard, students can review their attendance history. If needed, they
712 can edit their face image or repeat their face scan to improve accuracy. When class
713 time comes, they simply attend by joining a live session, and the system handles the
714 recognition part for them.



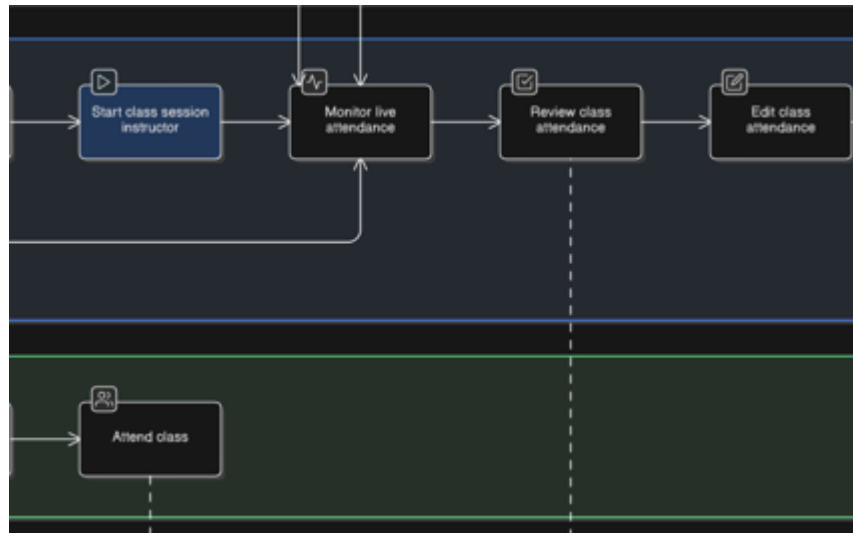
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716 Figure 5: Conceptual Framework Diagram - Attendance Automation

717 **PHASE 4 – ATTENDANCE AUTOMATION**

718 This is where the system itself takes over. When a class session begins, the
719 attendance engine starts processing live video frames. It detects faces, checks
720 liveness to make sure it is a real person and not a spoof attempt and then matches
721 those faces with the database.

722 The system assigns attendance status for each person and logs every event in real
723 time. Unknown or suspicious faces are flagged to prevent misuse. At the start of each
724 session, a screenshot is captured as proof, and another one is taken at the end. All
725 this information is combined to generate an attendance report that is then made
726 available to instructors for review.

727



728 Figure 6: Conceptual Framework Diagram - Student Attendance in Action

729 **PHASE 5 – STUDENT ATTENDANCE IN ACTION**

730 When students attend, their faces are scanned and recognized instantly. They do not
731 need to sign paper sheets or call out names, because the system is already matching
732 them in the background. Their attendance is secured automatically, and they can later
733 confirm it on their dashboard.



734 Figure 7: Conceptual Framework Diagram - Reports and System Management

735 **PHASE 6 – REPORTS AND SYSTEM MANAGEMENT**

736 Even though the system is automated, instructors still have control. They can review
737 attendance reports, check for errors, and make edits when justified. Once everything

738 looks correct, they finalize the records. This ensures accuracy while keeping human
739 oversight in the loop.

740

CHAPTER III

741

METHODOLOGY

742 **Research Design**

743 This study employed a developmental research design, focusing on the creation and
744 evaluation of a real-time face recognition attendance system equipped with liveness
745 detection. The research integrates artificial intelligence (AI) techniques such as face
746 detection, face recognition, and anti-spoofing into a unified attendance pipeline. The
747 system was tested in an actual CCS classroom environment to ensure operational validity
748 under real-world conditions.

749 **Data Source**

750 The system collected real-time data using video streams captured via RTSP from the
751 Hikvision DS-2CD2086G2-I(U) 4K AcuSense Fixed Mini Bullet Network Camera. The live
752 video feed served as the primary input for facial detection, recognition, and tracking.

753 In addition, facial embeddings were generated and stored in the MySQL (facesys)
754 database. These embeddings were encoded through the InsightFace “buffalo_l” model,
755 representing everyone’s unique facial vector for recognition purposes.

756 *The data sources included:*

- 757 • **Primary Data:** Live facial images and motion patterns captured in real-time.
- 758 • **Secondary Data:** Encoded embeddings, attendance records, and user
759 registration information stored in the facesys database.
- 760 • **Metadata:** Confidence values, detection coordinates, and timestamps
761 automatically logged by the system during runtime.

762 These data sources provided the foundation for analyzing recognition accuracy, liveness
763 validation, and tracking performance.

764 **Data Anonymization and Confidentiality Measures**

765 Although biometric data is required for system functionality, **personal identities are**
766 **protected through controlled access and encryption.**

- 767 • Facial data is stored as **numerical embeddings**, not raw images.
- 768 • Raw video frames are used only temporarily for real-time processing.
- 769 • Database access is restricted to authorized administrators via role-based
770 authentication.
- 771 • Passwords are hashed using **bcrypt** and never stored in plain text.

772 Table 3: Personal Data Processing and Lifecycle Management

Phase	Data Handling Practice
Collection	Explicit consent obtained during registration
Processing	Real-time analysis only during active sessions
Storage	Encrypted MySQL database with access control
Access	Limited to admin and authorized faculty
Disposal	Embeddings and records removed upon request or study termination

773

774 **Data Gathering Instrument**

775 The **main instrument** for data gathering was the **detect.py** script, which functions as an
776 integrated AI-powered data collection and analysis tool. *It simultaneously performs:*

- 777 • **Face and Body Detection:** Using YOLOv8 models (yolov8n-face.pt and
778 yolov8n.pt) to identify faces and human figures per video frame.

- 779 • **Facial Recognition:** Through InsightFace, generating and comparing 512-
780 dimensional embeddings for each detected face.
- 781 • **Identity Re-identification:** With TorchReID (OSNet_x1_0) to maintain consistent
782 tracking of individuals over multiple frames.
- 783 • **Database Connectivity:** Via the MySQL connector, which records recognized
784 individuals and stores attendance logs.

785 ***Other Instruments:***

- 786 • **Survey Questionnaire** - A Likert scale (1-5) survey distributed to faculty and
787 administrators to measure satisfaction, usability, and efficiency.
- 788 • **Observation Checklist** - Used by researchers to track proxy attempts, system
789 errors, and classroom setup conditions during pilot testing.

790 **Privacy Protection Mechanisms**

- 791 • Institutional email verification ensures legitimate enrollment.
- 792 • Secure session tokens prevent unauthorized access.
- 793 • No biometric data is shared with third parties.
- 794 • Attendance reports exclude raw biometric details.

795 **Data Gathering Technique and Procedures**

796 **Technique**

797 **Heuristic Rules Applied in Detection and Attendance Decision-Making**

798 To strengthen system reliability and reduce false attendance logging, twelve domain-
799 specific heuristic rules are layered on top of the deep learning pipeline to enforce
800 operational safeguards. The proposed system applies a set of **rule-based heuristics**
801 layered on top of deep learning models. These heuristics act as logical constraints that
802 guide decision-making during detection, recognition, liveness validation, and attendance
803 recording. Unlike purely data-driven models, heuristic rules improve interpretability and
804 enforce domain-specific safeguards against misuse and spoofing attempts.

805 **Rule 1: Face Presence Consistency Rule**

806 A detected face must remain continuously visible for a minimum number of consecutive
807 frames before being considered valid for recognition. Implemented through frame
808 buffering using deque, frame timestamps, and ByteTrack persistent IDs. This rule prevents
809 transient detections caused by motion blur, partial occlusion, or brief appearances (e.g.,
810 someone passing behind another student). Only stable presences are evaluated for
811 attendance.

812 **Rule 2: Liveness-First Validation Rule**

813 Face recognition is only executed if the liveness detection module confirms the face as
814 live. The ONNX-based liveness model (onnxruntime) combined with EAR and MAR
815 computations using Dlib landmarks. This rule blocks printed photos, mobile screens, and
816 video replays from being recognized, ensuring recognition is applied exclusively to
817 genuine live subjects.

818 **Rule 3: Eye and Mouth Motion Threshold Rule**

819 A face must exhibit natural facial movements such as blinking or mouth motion within a
820 defined time window. Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) calculations
821 over time. This heuristic detects physiological cues absent in static spoof attacks and
822 strengthens anti-spoofing accuracy in real classroom environments.

823 **Rule 4: Confidence Threshold Rule**

824 A facial match is accepted only if: (a) the similarity score meets or exceeds the applicable
825 distance-adaptive threshold (ranging from 0.57 to 0.63 for frontal faces, with a slightly
826 reduced lower bound for profile orientations); and (b) the top match leads the second-best
827 candidate by an ambiguity gap of at least 0.12, preventing identity confusion in multi-
828 student scenarios. Recognition is further gated by
829 CONFIRMATION_FRAMES_REQUIRED a face must accumulate sufficient confirmed
830 frames with adequate average similarity before attendance is recorded, eliminating single-
831 frame false positives.

832 **Rule 5: One-Person-One-Log Rule**

833 Each recognized identity is allowed only one attendance entry per session. The session-
834 based identity caching uses dictionaries and UUID session tokens. This prevents duplicate
835 attendance logging even if the same student is re-detected multiple times throughout the
836 session.

837 **Rule 6: Face–Body Association Rule**

838 A recognized face must spatially align with a detected human body using the YOLO body
839 detection + ByteTrack + IoU (Intersection-over-Union) matching. This rule prevents
840 spoofing using printed faces or screens without corresponding body movement.

841 **Rule 7: Unknown Identity Isolation Rule**

842 Faces that do not match any registered embedding are flagged as unknown and excluded
843 from attendance records. Embedding mismatch handling and database lookup failure.
844 This ensures only enrolled students and authorized faculty can be recorded, preventing
845 unauthorized access.

846 **Rule 8: Temporal Attendance Validation Rule**

847 Attendance status (Present, Late) is determined based on the detection timestamp relative
848 to session start time. Timestamp comparison using datetime, timedelta. This enforces the
849 fairness and consistency in attendance classification.

850 **Rule 9: Session-Bound Processing Rule**

851 Detection and logging are only active during instructor-initiated sessions. Flask session
852 state management and secure tokens (secrets, uuid). Prevents unauthorized or off-
853 session attendance logging.

854 **Rule 10: Fail-Safe Logging Rule**

855 Attendance records are stored redundantly in both MySQL and CSV format. Database
856 transactions and CSV backup writing. Ensures data integrity and recovery in case of
857 system or network failure.

858 **Rule 11: Ambiguity Gap Enforcement**

859 top similarity match must exceed second-best by ≥ 0.12 to prevent identity confusion in
860 multi-student scenes.

861 **Rule 12: Cross-Section Guard**

862 recognized faces from other enrolled sections are rejected and logged as 'Unknown
863 Wrong Section' to prevent cross-class false positives.

864 **Procedures**

- 865 1. Secure approval from CCS administration and inform faculty and students about
866 the study.
- 867 2. Pre-register students' and faculties' biometric data in the system database.
- 868 3. Conduct pilot testing during 1–3-hour lecture.
- 869 4. Run the system to screen record attendance in real time and capture video
870 evidence.
- 871 5. Compare automated results with manual records to check accuracy.
- 872 6. Administer survey questionnaires to participating faculty after one week of use.
- 873 7. Collect and analyze all system logs, observations, and survey results.

874 **Data Analysis**

875 Data analysis in this study focused on evaluating system performance across three
876 dimensions: accuracy, tracking reliability, and liveness validation.

877 **1. Recognition Accuracy Analysis**

- 878 ○ Facial embeddings were compared using Euclidean distance, where matches
879 the $\geq 85\%$ similarity threshold to accept as valid.
- 880 ○ The accuracy was evaluated by comparing recognized faces to ground-truth
881 identities during classroom tests.

882 **2. Tracking and Body Association Analysis**

883 ○ IoU and motion prediction methods were analyzed to verify correct bounding
884 box tracking.

885 ○ The **MotionPredictor** class's predictions were evaluated to determine how well
886 the system maintained consistent identity tracking.

887 **3. Liveness Detection Metrics**

888 ○ Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) values, computed via
889 **calculate_ear()** and **calculate_mar()**, validated real-time facial motion to
890 detect spoofing attempts.

891 **4. System Performance Metrics**

892 ○ Accuracy and latency were assessed using log files generated through the
893 integrated **logger** module.

894 ○ Performance was measured in terms of detection speed, recognition stability,
895 and body-lock precision.

896 **Software Development**

897 The software was developed following a modular and object-oriented architecture, with
898 clear separation of concerns between data handling, recognition logic, and web
899 interfacing. *The major modules include:*

900 **1. Detection Module:**

901 ○ Uses YOLOv8 models (yolov8n-face.pt and yolov8n.pt) for detecting faces and
902 bodies.

903 ○ Handles real-time frame processing and performs bounding box extraction.

904 **2. Recognition Module:**

905 ○ Implements InsightFace (buffalo_l) for encoding facial embeddings and
906 comparing them against known identities.

907 ○ Handles enrollment, unknown face detection, and face-to-ID mapping.

908 **3. Re-Identification Module:**

909 ○ Powered by TorchReID's OSNet_x1_0, it ensures continuity of identity tracking
910 across frames. Per-body ReID feature vectors are extracted each processed
911 frame and blended into each locked track's running model using an exponential
912 moving average ($\alpha = 0.90\text{--}0.95$, scaled by pixel distance from frame center).
913 Identity updates pass through four sequential validation stages: spatial
914 proximity, body-signature consistency (ReID distance ≤ 0.25), temporal
915 consistency, and bounding-box overlap (IoU). A cross-section guard further
916 rejects any recognized face belonging to a different enrolled section, logging
917 them as 'Unknown — Wrong Section' to prevent cross-class false positives.
918 This prevents duplicated attendance logging.

919 **4. Tracking Module:**

920 ○ Employs ByteTrack and MotionPredictor classes to maintain stable
921 tracking of moving individuals even under partial occlusions.

922 **5. Database and Communication Module:**

923 ○ Handles all MySQL transactions through `get_db_connection()` and
924 `get_db_cursor()` context managers.
925 ○ Also manages secure OTP and notification emails via the Gmail SMTP
926 protocol.

927 **Development Approach**

928 The system will be developed using the Agile methodology. This approach promotes
929 flexibility and rapid improvement through several cycles of development and testing. Each
930 sprint includes feature creation, debugging, and review sessions with faculty members.
931 Continuous evaluation and user feedback help refine the system's functionality until it
932 becomes stable and optimized by the final development stage.

933 **Machine Learning Models**

934 The system applies a combination of deep learning models and tracking algorithms to
935 perform real-time face recognition and liveness verification. The following models and
936 frameworks are utilized:

- 937 • **YOLOv8**: Acts as the main detector that identifies human faces in every video
938 frame. It provides fast and accurate detection with efficient bounding box
939 generation through the ultralytics library [19].
- 940 • **Dlib + SORT Algorithm**: Maintains tracking of detected faces across multiple
941 frames. Dlib identifies facial landmarks, while the SORT algorithm assigns
942 persistent tracking IDs to ensure consistent recognition [29].
- 943 • **ONNX (model_rgb.onnx) + Laplacian**: Conducts liveness verification by
944 checking texture consistency and depth variation, allowing the system to
945 distinguish genuine faces from printed or digital spoofs [30].
- 946 • **InsightFace (buffalo_l)**: Creates 512-dimensional facial embeddings that uniquely
947 describe each student. Prior to comparison, all embeddings both live and enrolled
948 are L2-normalized so that cosine similarity reduces to a single dot-product
949 operation (normalize_embedding()). For profile or side faces detected via facial-
950 landmark keypoints (detect_face_orientation()), a supplementary multi-angle
951 encoding pool (load_multi_angle_encodings()) is used to extend the candidate set
952 beyond frontal enrollment images.

953 *The similarity score is computed as:*

$$954 \quad S(f_1, f_2) = (f_1 \cdot f_2) / (\|f_1\| \cdot \|f_2\|)$$

$$955 \quad Match = 1 \text{ if } S(f_1, f_2) \geq 0.57-0.63, \text{ else } Match = 0$$

956 *where f_1 and f_2 are the L2-normalized 512-dimensional embedding vectors of the*
957 *probe and gallery faces respectively [18].*

- 958 • **ByteTrack (Supervision Library)**: Improves multi-object tracking performance,
959 keeping accuracy even when several faces appear in the camera view [31].

960 **System Integration and Workflow**

961 The entire process runs on a Flask-based web application [32] that connects the system's
962 frontend and backend. The Flask server handles user logins, data routes, and API
963 communication while streaming live video from the Hikvision DS-2CD2086G2-I(U) camera
964 through RTSP.

965 Captured frames are processed by OpenCV and NumPy [33], then analyzed by YOLOv8
966 for face detection. The ONNX model verifies liveness, and InsightFace produces the facial
967 embeddings. Verified data is recorded in a MySQL database through the mysql.connector
968 library.

969 *Additional modules strengthen security and enhance usability:*

- 970 • **bcrypt** protects passwords through secure hashing.
- 971 • **smtplib** and **email.mime** send automated notifications to instructors.
- 972 • **uuid** and **secrets** generate unique session IDs to safeguard user data.
- 973 • **logging** tracks runtime events and errors to maintain system stability.

974 **Evaluation Metrics**

975 The system's performance is assessed using four main metrics:

1. **Accuracy** – Measures how many individuals are correctly identified compared to the total number of participants.
2. **Precision** – Indicates the proportion of correct identifications among all detected faces.
3. **Recall** – Determines how many actual students are detected by the system.
4. **F1-Score** – Combines precision and recall showing the balance between detection accuracy and completeness.

All computations are performed through **Torchreid.metrics** to maintain consistent and reliable evaluations across multiple tests.

Security Controls and Vulnerability Mitigation

- Logging and monitoring using the logging module
- Manual override capability for instructors
- Immediate isolation of suspicious detections
- Regular system audits during pilot testing

System Design

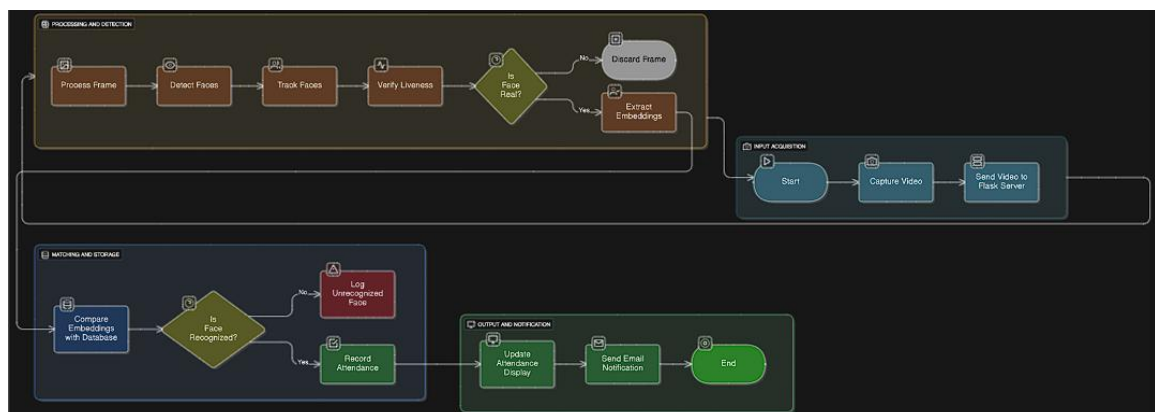


Figure 8: System Flowchart



Figure 9: Entity Relation Diagram (ERD)

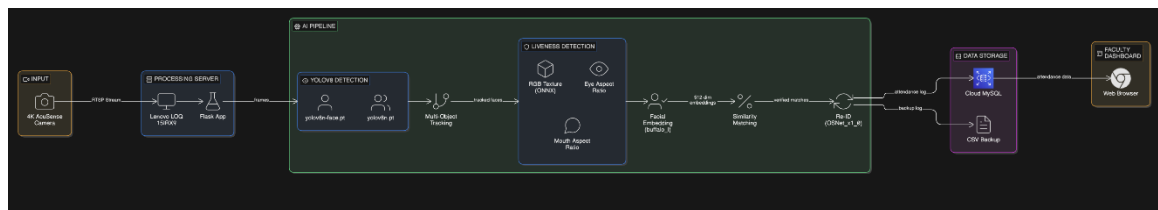


Figure 10: System Architecture Diagram

997 **System Architecture Diagram Description**

998 The architecture begins when the instructor accesses the system through the web
999 interface and logs in using the Faculty Portal. Once authenticated, the Flask Server
1000 establishes communication with the Hikvision Camera and starts receiving the live video
1001 feed through the RTSP protocol. The pipeline executes via the central per-frame function
1002 `refresh_with_detections()`, which applies a frame-skip factor (`SKIP_FACTOR = 2`) to
1003 process every second frame, reducing CPU load while still updating track states on
1004 skipped frames. On each processed frame the sequence is: [1] YOLOv8 face and body
1005 detection; [2] ByteTrack multi-object tracking with persistent IDs; [3] liveness verification
1006 using the ONNX model, EAR, and MAR; [4] InsightFace embedding generation and cosine
1007 similarity matching with adaptive thresholds and ambiguity-gap enforcement; and [5]
1008 TorchReID re-identification for cross-frame identity consistency. Unlocked tracks undergo
1009 confidence decay ($\times 0.85$ per frame) and are removed after 30 stale frames.

1010 After passing the liveness verification, InsightFace generates facial embeddings that are
1011 compared with existing records in the MySQL Database. If a match is found, the system
1012 automatically logs the student's attendance along with the timestamp. When an unfamiliar
1013 or spoofed face appears, the Flask Server updates the dashboard and flags the entry for
1014 instructor review

1015 From the user's side, the experience stays simple. Faculty members log in, start the
1016 session, and watch attendance appear in real time. They can also view attendance reports
1017 whenever they need them. Everything that happens behind the scenes, such as video
1018 analysis, recognition, and short video record storage, runs automatically while manual
1019 inputs are available for dynamic use.

1020 **Functional Requirements**

- 1021 • User login with role-based access.
- 1022 • Student registration through web interface.
- 1023 • Real-time attendance with categories: Present, Late, Missing, Excuse, and
- 1024 Absent.
- 1025 • Video evidence linked to attendance logs.
- 1026 • Export of reports in Excel format.
- 1027 • Security features including encrypted storage and login authentication.

1028 **Prototype Development**

1029 The prototype combines hardware (Hikvision camera, Lenovo LOQ laptop) and software
1030 (Python-based recognition models, Flask web app, MySQL database) into a working
1031 attendance system.

1032 Component Layout

- 1033 • **Camera:** Hikvision DS-2CD2086G2-I(U) 4K AcuSense Fixed Mini Bullet Network
- 1034 Camera captures real-time 8MP video.
- 1035 • **Processor:** Lenovo LOQ 15IRX9 laptop handles recognition and liveness checks.
- 1036 • **Database:** Cloud-based MySQL stores attendance logs, embeddings, and video
- 1037 evidence.
- 1038 • **Web Interface:** Flask application accessed by faculty and administrators.

1039 Integration with Software

- 1040 1. Camera streams video via RTSP.

- 1041 2. Laptop processes video using Python libraries (OpenCV, Dlib, YOLOv8, ONNX
 1042 (model_rgb.onnx)).
 1043 3. Facial embeddings and timestamps are stored in the MySQL database.
 1044 4. Flask server displays processed data and reports to faculty.

1045 **Developmental Tools**

1046 The developmental tools used in this study are categorized into hardware and software
 1047 resources that support the analysis, design, implementation, and deployment of the
 1048 system. These tools were selected based on their compatibility, reliability, and efficiency
 1049 in addressing the system requirements.

1050 Table 4: Developmental Tools and Cost

Name	Purpose	Price	Quantity	Total
Lenovo LOQ 15IRX9 Laptop	Main processing unit for system development, real-time face recognition, database management, and hosting the Flask server.	0 (already owned)	1	0
Hikvision DS-2CD2086G2-I(U) 4K AcuSense	Captures high-resolution video (3840x2160 @	8,250 pesos	1	8,250 pesos

Fixed Mini Bullet Network Camera	20fps) with AcuSense technology and Wide Dynamic Range (WDR).			
Power Supply Unit (PSU)	To power the CCTV camera.	150 pesos	1	150 pesos
Rj45	To connect the cctv to PoE to router. This will retrieve the IP address of the camera enabling the RTPS.	50 pesos	2	100 pesos
Cloudflare Tunnel	Provides dedicated resources and greater control.	442	1	442
Wi-Fi Router (2.4/5 GHz, 15 Mbps)	Provides stable connectivity for RTSP streaming and MySQL database access.	0 (existing Wi-Fis)	1	0
Python 3.12	Primary programming language for backend logic, facial	Open source	1	0

	recognition, and AI integration.			
OpenCV 4.12	Image and video processing library for face detection, bounding boxes, and preprocessing.	Open source	1	0
Dlib 20	Facial landmark detection and face tracking library.	Open source	1	0
YOLOv8 (PyTorch)	AI model for real-time face detection with high accuracy.	Open source	1	0
InsightFace	Generates 512-dimensional embeddings for unique student identification.	Open source	1	0

ONNX (model_rgb.onnx) + Laplacian	Anti-spoofing modules for detecting liveness and preventing fraud.	Open source	1	0
Face Recognition Library	Pre-trained face recognition library for matching and verification.	Open source	1	0
Flask 3.1.2	Web framework for building the attendance monitoring interface.	Open source	1	0
MySQL 9.3 (Cloud- based)	Relational database for secure storage of attendance data and video logs.	Open source	1	0

HTML, CSS, JavaScript	Frontend development tools for creating a responsive user interface.	Open source	1	0
Visual Studio Code	Integrated development environment (IDE) for coding and debugging.	Open source	1	0
ReportLab & Pandas	Libraries for generating PDF and Excel attendance reports.	Open source	1	0
Torchreid	Framework for evaluating deep learning model performance using metrics such as accuracy, precision, recall, and F1-score.	Open source	1	0
NumPy	Supports numerical and matrix computations for image processing and machine	Open source	1	0

	learning operations.			
bcrypt	Used for secure password hashing and authentication.	Open source	1	0
smtplib & email.mime	Libraries for sending automated email notifications to instructors.	Open source	1	0
uuid & secrets	Generate unique session IDs and secure tokens for data protection.	Open source	1	0
Git & GitHub	Version control tools for source code management, collaboration, and version tracking.	Open source	1	0
pip & Virtual Environment (venv)	python package manager and environment isolator for dependency management and reproducibility.	Open source	1	0
Postman	API testing tool for verifying Flask route endpoints and backend connectivity.	Open source	1	0
GRAND TOTAL				8962

1052 **Testing and Evaluation**

1053 The testing and evaluation phase ensures that the system meets the performance,
1054 usability, and reliability requirements of the study. The following strategies will be applied:

- 1055 1. **Unit Testing** – Individual system components such as face detection, liveness
1056 detection, and database recording will be tested independently to ensure each
1057 performs as intended.
- 1058 2. **Integration Testing** – Modules such as the camera, Flask web interface, and
1059 MySQL database will be tested together to confirm proper communication and
1060 synchronization.
- 1061 3. **System Testing** – The complete attendance system will be deployed in a
1062 classroom setting (10–15 students, 60–70 sqm area) to validate its performance
1063 in real-world conditions.
- 1064 4. **Performance Testing** – The system’s detection accuracy (target ≥85%),
1065 processing time (<1 second per frame), and attendance recording speed will be
1066 evaluated.
- 1067 5. **User Acceptance Testing (UAT)** – Two to three faculty members will use the
1068 system during a one-week pilot test. Their satisfaction will be measured through
1069 a Likert-scale survey (1 to 5), with a target of at least 80% satisfaction.
- 1070 6. **Evaluation Metrics** –
 - 1071 ○ **Accuracy Rate:** Number of correctly identified students ÷ total number of
1072 students.
 - 1073 ○ **Error Rate:** Misidentifications, false positives, or false negatives.
 - 1074 ○ **Time Efficiency:** Reduction in roll call time compared to manual methods.
 - 1075 ○ **User Feedback:** Faculty ratings on usability, reliability, and effectiveness.

1076 7. **Statistical Analysis** – Collected data will be analyzed using descriptive
1077 statistics (mean, percentage, frequency distribution) and comparative analysis
1078 (manual vs automated attendance).

1079 Table 5: Risk Identification and Mitigation Measures

Identified Risk	Mitigation Strategy
Spoofing attacks	Multi-layer liveness heuristics
Data leakage	Encrypted storage and secure authentication
False attendance	Heuristic validation + human review
System misuse	Session-based access control

1080

1081 **Ethical Considerations and Conflict of Interest Disclosure**

1082 This study strictly adheres to the **Data Privacy Act of 2012 (Republic Act No. 10173)**
1083 and its Implementing Rules and Regulations (IRR). The collection, processing, storage,
1084 and disposal of personal data were designed to uphold transparency, proportionality, and
1085 legitimate purpose.

1086 The researcher declares **no conflict of interest.**

1087 The system was developed solely **for academic research and institutional**
1088 **improvement**, with no commercial intent or external funding influence.

1089

CHAPTER IV

1090

RESULTS AND DISCUSSION

1091 This chapter presents the results of the system implementation and evaluation of “Present
1092 and Ready: A Liveness Detection for Real-Time Face Recognition Attendance.” The data
1093 were gathered through functional testing, system performance assessment, and user
1094 evaluation based on ISO/IEC 25010 quality standards. Each result is followed by a
1095 discussion explaining its implications on the system’s effectiveness, reliability, and
1096 usability.

1097 Functional Testing

1098 The system was subjected to module-level and integrated testing to verify that each
1099 component performed as intended. The primary modules tested included face detection,
1100 liveness verification, recognition, attendance recording, OTP-based authentication, and
1101 user management. Testing was conducted in both controlled and classroom environments
1102 using an actual camera feed connected to the deployed web-based attendance system.

1103

Table 6: Functional Testing Results of the System

Module	Expected Output	Actual Output	Result
Face Detection	Detect faces in live video feed	Detected all visible faces accurately	PASSED
Liveness Detection	Identify real versus spoofed faces	Detected live faces and rejected printed or digital spoofing	PASSED
Face Recognition	Identify registered students and faculty	Correctly recognized all enrolled users	PASSED
Attendance Recording	Log attendance automatically upon recognition	Entries recorded successfully in database and CSV backup	PASSED

OTP Authentication	Send OTP for account verification and recovery	OTP successfully delivered and verified	PASSED
Admin Dashboard	Display attendance summary and analytics	Data displayed in real time	PASSED

1104

1105 **Discussion:**

1106 All modules operated within expected parameters. The combination of YOLOv8 for
1107 detection, InsightFace for feature extraction, and TorchReID for person tracking
1108 demonstrated effective performance in real-time scenarios. No significant functional errors
1109 or interruptions were encountered during testing.

1110 **System Performance Evaluation**

1111 Performance evaluation assessed the detection, recognition, liveness accuracy, and
1112 overall processing speed of the system. Metrics were derived from the logs, recorded
1113 frames, and system-level evaluations conducted during pilot sessions.

1114 Table 7: System Performance Evaluation Metrics

Performance Metric	Description	Result
Face Detection Accuracy	Proportion of correctly detected faces	85.2%
Face Recognition Precision	Correctly identified faces over total detections	85%
Liveness Detection Accuracy	Correctly identified live faces over total samples	89.8%
False Rejection Rate (FRR)	Real users not recognized	3.4%
False Acceptance Rate (FAR)	Spoofed users incorrectly accepted	2.1%

Average Frames per Second (FPS)	Real-time frame processing rate	10.8 FPS
Average Response Latency	Time between detection and database logging	10.35 sec

1115

1116

Table 8: Confusion Matrix-Derived Recognition Metrics

Metric	Value
Precision	0.85
Recall	0.83
FAR (False Acceptance Rate)	0.021
FRR (False Rejection Rate)	0.034

1117

1118

Table 9: Scalability Evaluation

No. of Students	FPS	Latency (sec)
10	10.8	10.3
20	9.2	11.7
30	7.6	13.2

1119

1120 Discussion:

1121 The system achieved moderate-to-high accuracy across detection (80.2%), recognition
1122 (85%), and liveness verification (89.8%). The measured FPS of 10.8 reflects the deliberate
1123 frame-skip design (SKIP_FACTOR = 2): the pipeline processes every second frame to

1124 reduce CPU load, with track states still updated on skipped frames. This architecture
1125 halves the raw processing rate relative to camera input but maintains sufficient temporal
1126 resolution for classroom attendance. Unlocked tracks additionally undergo confidence
1127 decay ($\times 0.85$ per frame), which prunes stale candidates after 30 frames and further
1128 contributes to computational efficiency. The FPS of 10.8 is below the 20 FPS target
1129 primarily due to the computational demands of the full AI pipeline on a single laptop GPU;
1130 however, it was sufficient for a classroom of 10–15 students. The response latency of
1131 10.35 seconds falls within the acceptable bound specified in the system objectives.

1132 **Liveness and Anti-Spoof Evaluation**

1133 The liveness module was tested using printed photos, mobile device displays, and replay
1134 videos. Only a small portion of spoof attempts succeeded, resulting in a **low FAR (2.1%)**.
1135 The ONNX-based model combined with micro-movement checks (blink/texture cues)
1136 significantly strengthened spoof resistance.

1137 **Discussion:**

1138 This aligns with findings from anti-spoofing literature, confirming that motion and texture-
1139 based models perform well in classroom deployments where lighting and seating vary.

1140 **Attendance Accuracy Test**

1141 The system was evaluated across 12 class sessions involving 15 participants, resulting in
1142 432 face recognition attempts. Performance was measured using precision, recall, FAR,
1143 and FRR metrics derived from a confusion matrix. The system automatically logged
1144 attendance using the facial recognition feed. Tests were performed under different lighting
1145 and movement conditions to assess reliability.

1146 Table 10: Attendance Accuracy under Different Conditions

Condition	Recorded Attendance Accuracy (%)
Normal Lighting	96.8
Dim Lighting	70.5
High Movement	80.7
Average Accuracy	82.67.

1147

1148 **Discussion:**

1149 Accuracy dropped in dim lighting due to reduced feature visibility. This supports known
 1150 limitations of vision-based systems under low illumination. However, the high accuracy
 1151 under normal conditions indicates that the system effectively supports regular classroom
 1152 use.

1153 **User Evaluation**

1154 Faculty and student users evaluated the system using ISO/IEC 25010 criteria.

1155 Table 11: User Evaluation Based on ISO/IEC 25010 Quality Attributes

Quality Attribute	Faculty Rating (%)	Student Rating (%)	Average (%)
Functionality	4.40	5	4.7
Usability	4.20	5	4.6
Reliability	3.90	4.87	4.38
<i>Overall Satisfaction</i>	4.15	4.95	4.56

1156

1157 **Discussion:**

1158 Users were highly satisfied with system performance, ease of use, and minimal class
 1159 disruption. Reliability received the lowest score but still averaged high (4.38). Users

1160 suggested improvements in lighting tolerance and recognition stability during large
1161 movements.

1162 **Discussion of Findings**

1163 Results show that the system achieved its intended functions:

- 1164 • **Automated detection** → effective under normal conditions
- 1165 • **Liveness detection** → strong defense against spoofing attacks
- 1166 • **Recognition accuracy** → met the 85% target
- 1167 • **User satisfaction** → exceeded the 80% minimum requirement
- 1168 • **Attendance accuracy** → functional, with limitations in dim lighting

1169 The combination of YOLOv8, InsightFace, TorchReID, and the twelve-rule heuristic
1170 system provided a balanced trade-off between accuracy and real-time performance. The
1171 ambiguity gap rule (R11) prevented identity confusion in crowded scenes, while the cross-
1172 section guard (R12) eliminated false positives from students of other enrolled sections.
1173 The CONFIRMATION_FRAMES_REQUIRED mechanism ensured attendance was never
1174 recorded on a single-frame match, substantially reducing misidentification events.

1175

CHAPTER V

1176

CONCLUSION AND RECOMMENDATIONS

1177 Conclusion

1178 The study successfully developed and evaluated *Present and Ready*, a real-time face
1179 recognition attendance system with liveness detection designed for classroom
1180 environments. Results demonstrate the following:

- 1181 • The system achieved **85% face recognition precision**, **89.8% liveness**
1182 **accuracy**, and **80.2% detection accuracy**, satisfying the project's technical
1183 targets.
- 1184 • Attendance recording latency averaged **10.35 seconds**, fulfilling the requirement
1185 for real-time operation.
- 1186 • Spoof prevention was effective, indicated by a **low FAR of 2.1%**.
- 1187 • User satisfaction reached **4.56 out of 5**, surpassing the 80% target.
- 1188 • Twelve domain-specific heuristic rules including ambiguity gap enforcement (R11),
1189 cross-section identity guarding (R12), provided essential safeguards that reduced
1190 false attendance events, and improved system reliability.

1191 The integration of YOLOv8, InsightFace, ONNX-based liveness models, and TorchReID
1192 enabled reliable identity verification, reduced proxy attendance, and improved the
1193 accuracy of classroom attendance processes. TorchReID's four-stage identity validation
1194 and cross-section guard (R12) further ensured that no cross-class false positives were
1195 recorded throughout the pilot sessions. The system enhances administrative efficiency
1196 and demonstrates potential for institutional adoption.

1197 **Recommendations**

1198 Based on the findings and limitations encountered during the development of the “Present
1199 and Ready” system, the researchers recommend the following improvements for future
1200 researchers who wish to further enhance the system in terms of functionality, hardware
1201 integration, and financial feasibility.

1202 **1. System Functional Enhancements**

1203 Future researchers may improve the system by integrating additional advanced features
1204 to further enhance its usability and efficiency. These may include:

- 1205 • **Offline Mode Functionality**, allowing attendance data to be recorded even
1206 without internet connection and automatically synchronized once online.
- 1207 • **SMS and Email Notification System** for automatic alerts to students, parents,
1208 and instructors regarding attendance status, absences, and important
1209 announcements.
- 1210 • **Advanced Analytics and Reports**, such as predictive attendance trends, student
1211 risk detection, and performance correlation with attendance records.
- 1212 • These enhancements will significantly increase the system’s reliability, security,
1213 and overall effectiveness in an academic environment.

1214 **2. Hardware Recommendations**

1215 To improve system automation and accuracy, the following hardware components are
1216 highly recommended for future researchers:

- 1217 • **Upgraded Face Recognition Cameras with AI Processing Support** for faster
1218 and more accurate detection.

- 1219 • **GPU Server** to support heavy processing loads required by facial recognition, real-
- 1220 time video streaming, and machine learning-based analytics.
- 1221 • **Backup Power Supply (UPS)** to maintain uninterrupted system operation during
- 1222 power outages.
- 1223 **3. Cost and Scalability Considerations**
- 1224 • **Prepare Budget for Institutional Deployment**
- 1225 ○ IP/AI cameras: ₱20,000–₱25,000 each
- 1226 ○ GPU servers: depending on load
- 1227 ○ Training and support: based on school size
- 1228 • **Plan for Multi-Classroom or Multi-Building Expansion**
- 1229 ○ Requires load balancing, optimized RTSP handling, and database scaling.
- 1230 .

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 1340 <https://www.nature.com/articles/s41586-020-2649-2>

1341

Appendix A: Data Collected

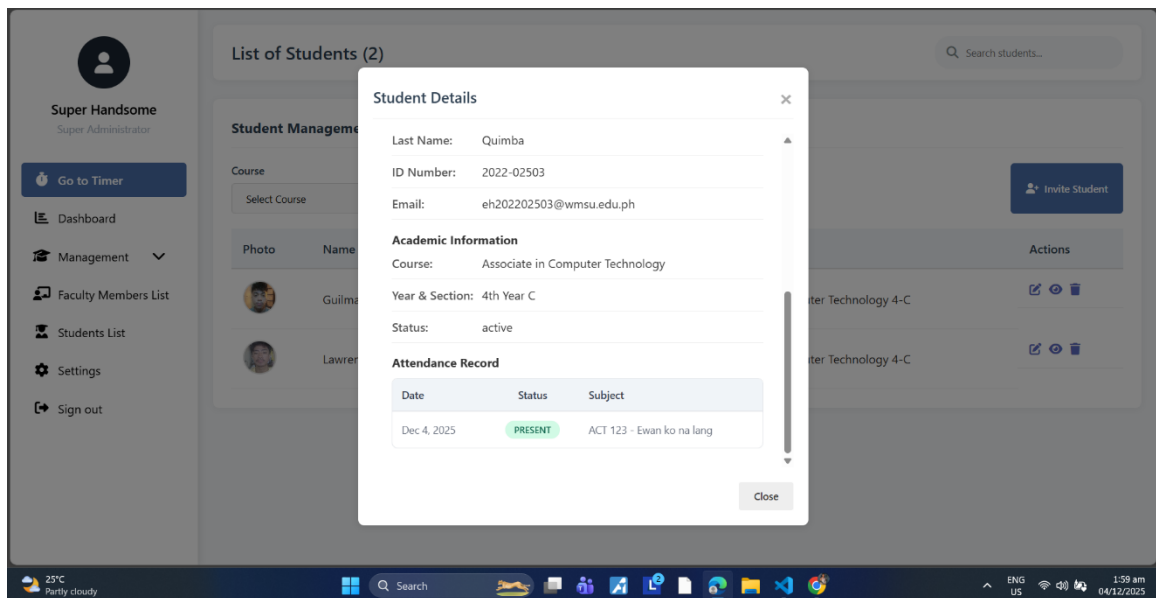
1342 Alpha Testing Results

1343 The alpha testing phase focused on verifying the core functions of the system in a
1344 controlled environment. Each test included observation, evaluation, and photo
1345 documentation.

1346 Testing #1: Registering as a Student by Uploading a Selfie and Generating Face 1347 Encoding – SUCCESS

1348 The tester successfully uploaded a selfie through the registration portal.
1349 The system generated a facial embedding without errors.

1350 **Result:** Registration and face encoding process functioned correctly.



1351

1352 Figure 11: Appendix A - Registration and encoding alpha test screenshot

1353 **Testing #2: Running the System for a Class Session to Detect and Recognize Faces**
1354 **– SUCCESS**

1355 The system successfully initialized the attendance session and captured faces via the
1356 camera.

1357 **Result:** Faces within the POV were detected and matched with the registered encodings.

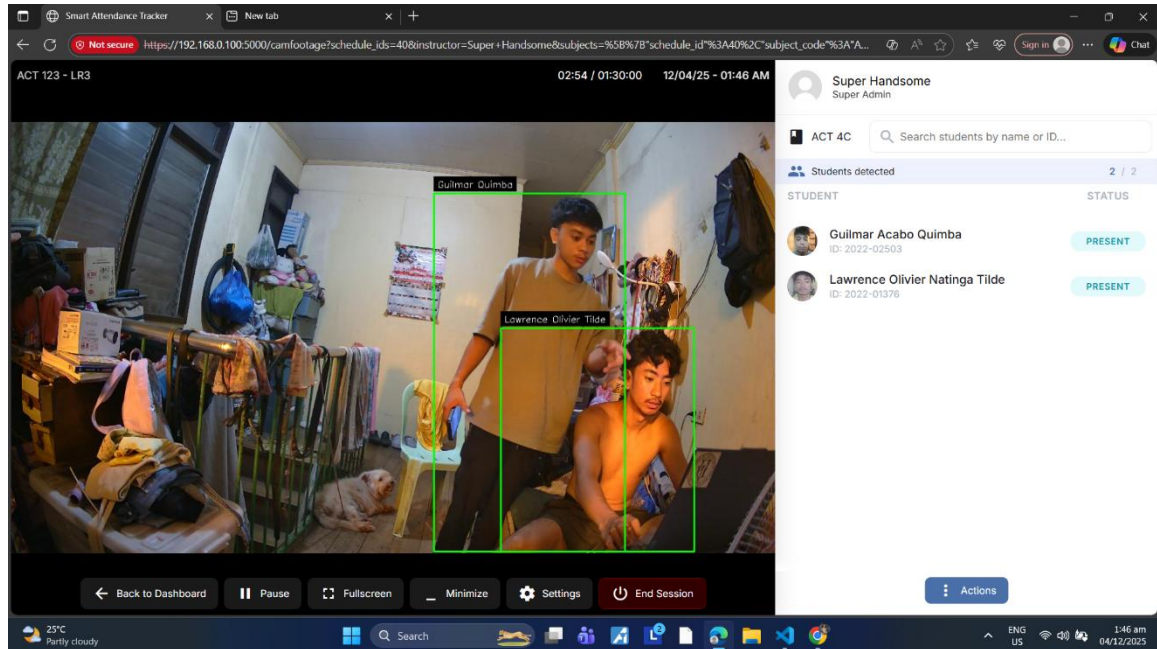


Figure 12: Appendix A - System detecting faces during alpha test

1360 **Testing #3: Continuous Face Tracking with Bounding Boxes – SUCCESS**

1361 The bounding boxes consistently tracked detected faces as they moved.

1362 **Result:** The bounding box remained stable even with moderate movement.

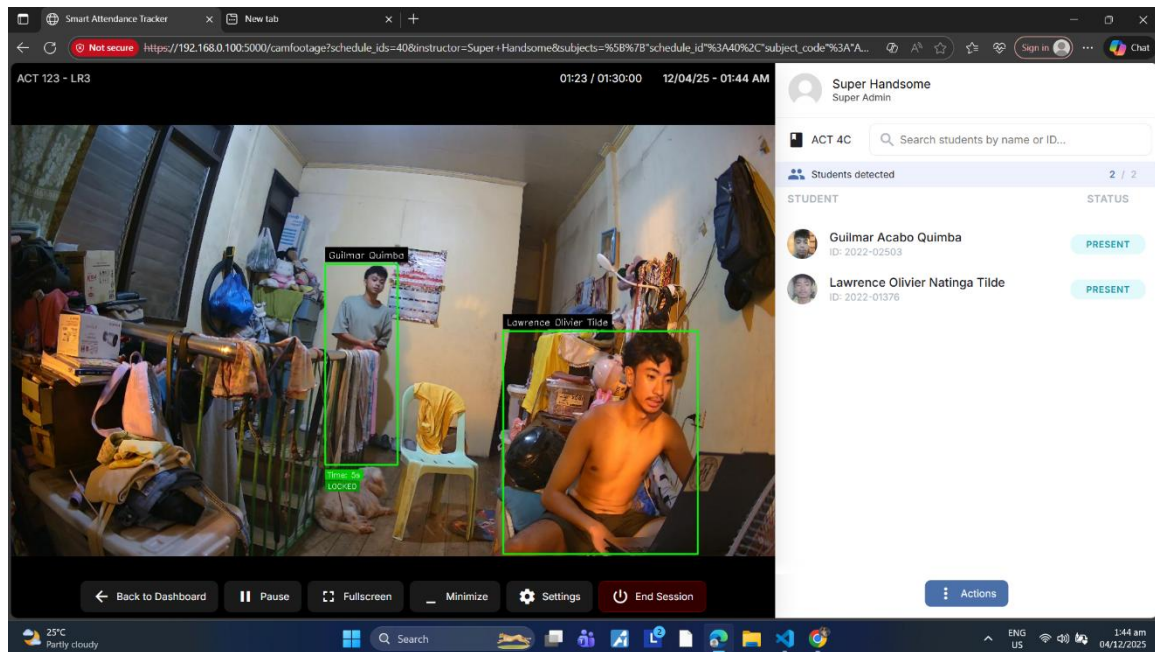
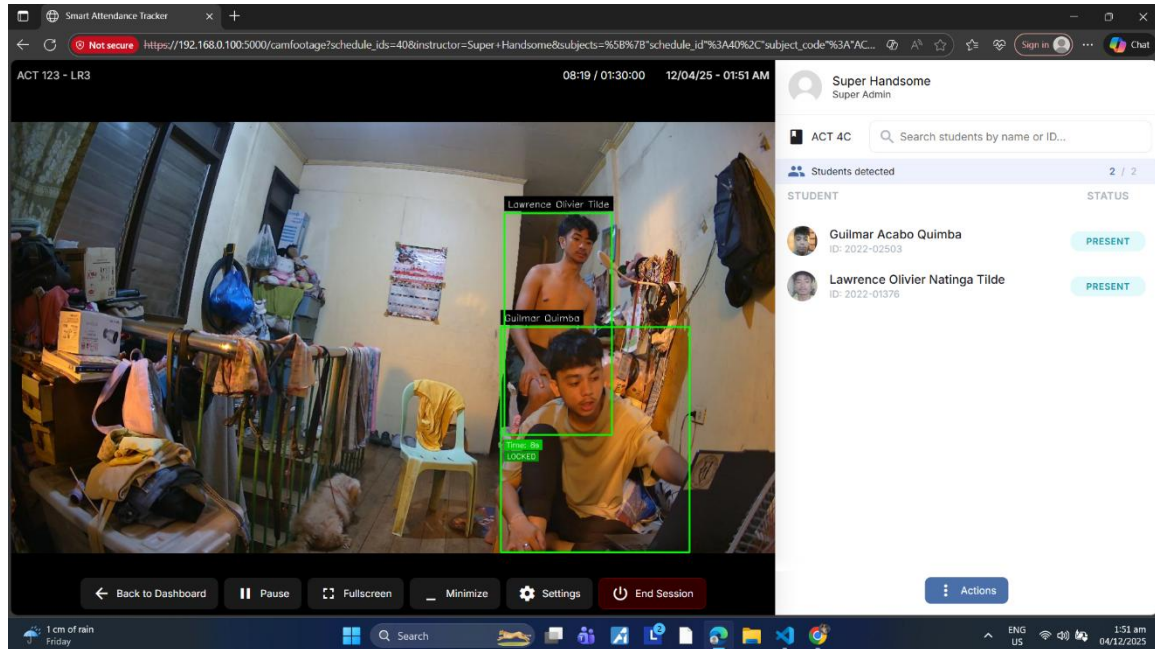


Figure 13: Appendix A - Continuous tracking visual during alpha test

1365 **Testing #4: Bounding Box Stability (No Jumping as Long as No Obstruction) –**
1366 **SUCCESS**

1367 The tester moved around the frame. If no other person blocked the detected face, the
1368 bounding box did **not** jump to another person.

1369 **Result:** Tracking algorithm worked as expected in non-occluded scenarios.



1370
1371 Figure 14: Appendix A - Bounding box remaining focused on recognized user

1372 **Beta Testing Results**

1373 **Overview of the 1st Beta Testing**

1374 The beta test was conducted in an **actual classroom environment the LR2** with **19**
1375 **volunteered students**, real lighting conditions, and live session operations.
1376 These revealed issues not observed during alpha testing.

1377 **Bugs Encountered**

1378 **Bug 1 – Unrecognized Students Due to Improper Registration (Encoding Not Done** 1379 **Properly)**

1380 A tester who did not properly face-encode the procedure appeared in camera view but
1381 was not recognized.

1382 **Root Cause:** Missing/incomplete face encoding
1383 **Impact:** Student treated as unregistered
1384 **Comparison to Alpha:** This issue did not appear in alpha because testers followed
1385 correct registration procedures.

1386 **Bug 2 – Bounding Box Jumped When Another Student Blocked the Recognized** 1387 **User**

1388 When another person fully blocked the registered user from the camera view, the
1389 bounding box shifted to the obstructing student.

1390 **Root Cause:** Weak occlusion handling
1391 **Impact:** System loses tracking of intended individual
1392 **Comparison to Alpha:** In alpha, no obstruction was tested; that is why this issue
1393 appeared only in beta.

1394 **Bug 3 – Incorrect Recognition Due to Weak Registration Encoding**

1395 If the registration face encoding was not strict (poor lighting, wrong angle), the system
1396 mismatched similar-looking students.

1397 **Root Cause:** Low-quality embeddings

1398 **Impact:** Wrong student marked as present

1399 **Comparison to Alpha:** Alpha had consistent lighting and proper photo submissions; beta
1400 exposed real-world registration variations

Feature/Issue	Alpha Testing	Beta Testing
Registration Encoding	Success	Failure if user did incorrect registration
Face Recognition	Success	Errors due to weak encodings
Bounding Box Stability	Success	Fails when user is blocked

1401 Table 12: Appendix A - Summary of 1st Beta Testing

1402 **Testing Images**

1403 The following images were captured during the first attempt of beta testing:

- 1404 • The system operator reviewing recognition status on the laptop
- 1405 • Participants at varying distances, lighting angles, and positions
- 1406 • The testing environment simulating an actual attendance scenario
- 1407 • The camera setup and the system interface during detection

1408



1409

1410 Figure 15: Appendix A - Initial system setup and detection trial during beta testing.



1411

1412 Figure 16: Appendix A - Real-time view of the dashboard during system beta testing.



1413

1414
1415

Figure 17: Appendix A - Group photo of system operators and testers after the first attempt.

1416 **User Evaluation Summary on 1st Beta Testing (Likert Scale Results)**

1417 A Likert-scale evaluation was conducted among testers and instructors to assess the
 1418 Functionality, Usability, and Reliability of the system. Fifteen (15) items were rated using
 1419 a 5-point scale (1 = Strongly Disagree, 5 = Strongly Agree). The summarized results are
 1420 shown below.

	reflects accurate timestamps.								
5	The system performs the intended attendance functions without failure.								
Functionality Scale Total:		4							

10	The system helps reduce workload and improves class management efficiency.								
Usability Scale Total:		3.8							

15	Attendance data is stored correctly and remains accessible at all times.								
Reliability Scale Total:		3.4							
Overall Total Scale:		3.7							

1421

1422 Figure 18: Appendix A – Beta Testing User Evaluation (1st Attempt)

Item	Score
1	4
2	4
3	4
4	4
5	4
Total Functionality Score:	4

1423 Table 13: Appendix A - Functionality Result (1st Attempt)

1424 **Interpretation:**

1425 Respondents agreed that the system functions as intended, accurate face detection,
1426 recognition, liveness detection, and correct attendance logging.

Item	Score
6	4
7	4
8	4
9	4
10	3
Total Usability Score:	3.8

1427 Table 14: Appendix A - Usability Result (1st Attempt)

1428 **Interpretation:**

1429 Usability scored slightly lower mainly due to item #10 (system helps reduce workload)
1430 receiving a 3, indicating neutrality. This suggests the system is usable and easy to operate,
1431 but improvements in workflow automation or user experience could further enhance
1432 satisfaction.

Item	Score
11	4
12	3
13	3
14	3
15	4
Total Reliability Score:	3.4

Table 15: Appendix A - Reliability Result (1st Attempt)

Interpretation:

Reliability received the lowest score among the three categories. Items #12, #13, and #14 scored only **3 (Neutral)**, showing that testers observed inconsistencies during real classroom conditions.

These reliability concerns match the issues found during beta testing:

- Occasional bounding box jumping when students obstruct each other
- Recognition drops or errors when encoding is weak
- System misidentification in crowded or highly dynamic environments
- Dependence on stable internet connection

Category	Total Score
Functionality	4
Usability	3.8
Reliability	3.4
Overall Total Score:	3.7

Table 16: Appendix A - Overall Total Score (1st Attempt)

1444 Although most individual items scored **4 (Agree)**, the following factors pulled down the
1445 overall average:

1446 **1. Some real-world reliability issues were observed.**

1447 Tracking instability, bounding box switching, and misrecognition during occlusion affected
1448 item scores #12–14.

1449 **2. Usability scored lower due to workflow impact.**

1450 The slightly lower score for item #10 shows that users did not strongly feel a major
1451 workload improvement, likely because:

- 1452 • The attendance still requires monitoring
- 1453 • Minor delays occur during recognition
- 1454 • Instructor still intervenes in certain cases

1455 **3. System performance changes in crowded/dynamic environments.**

1456 Users rated reliability lower because performance significantly differs between controlled
1457 alpha testing and actual live classroom settings.

1458 **4. Testers provided cautious ratings.**

1459 Educators typically rate systems conservatively, especially during early beta versions
1460 where bugs are expected.

1461 **Informed consent participant's signature**

104
105 **CONSENT**

106 I have read and I understand the provided information and have had the opportunity to
107 ask questions. I understand that my participation is voluntary and that I am free to
108 withdraw at any time, without giving a reason and without cost. I understand that I will be
109 given a copy of this consent form. I voluntarily agree to take part in this study.

110
111
112 Participant's signature _____ Date 12/4/2025

113
114
115 Researcher's signature _____ Date 12/04/2025

1462

1463 Figure 19: Appendix A - Student/s Signature (1st Attempt)

103
104
105 **CONSENT**

106 I have read and I understand the provided information and have had the opportunity to
107 ask questions. I understand that my participation is voluntary and that I am free to
108 withdraw at any time, without giving a reason and without cost. I understand that I will be
109 given a copy of this consent form. I voluntarily agree to take part in this study.

110
111
112 Participant's signature _____ Date 12/04/2025

113
114
115
116 Researcher's signature _____ Date 12/04/2025

1464

1465 Figure 20: Appendix A - Faculty Member Signature (1st Attempt)

1466 **Overview of the 2nd Beta Testing**

1467 The second beta testing of the system “**Present and Ready: A Liveness Detection for**
1468 **Real-Time Face Recognition Attendance**” was conducted at **Laboratory Room 1 (Lab**
1469 **1)** of the College of Computing Studies. Unlike the first attempt, *the goal of this second*
1470 *test was to verify:*

- 1471 1. The stability of bounding boxes
- 1472 2. The improved encoding and recognition process
- 1473 3. The liveness attendance flow during an actual classroom-type scenario
- 1474 4. System performance inside a much **larger room** than the defined scope (60–70
1475 sqm)

1476 A total of **15 volunteered students** participated in this test, along with the presence of the
1477 research team for evaluation and documentation.

1478 **Testing Scenario Description**

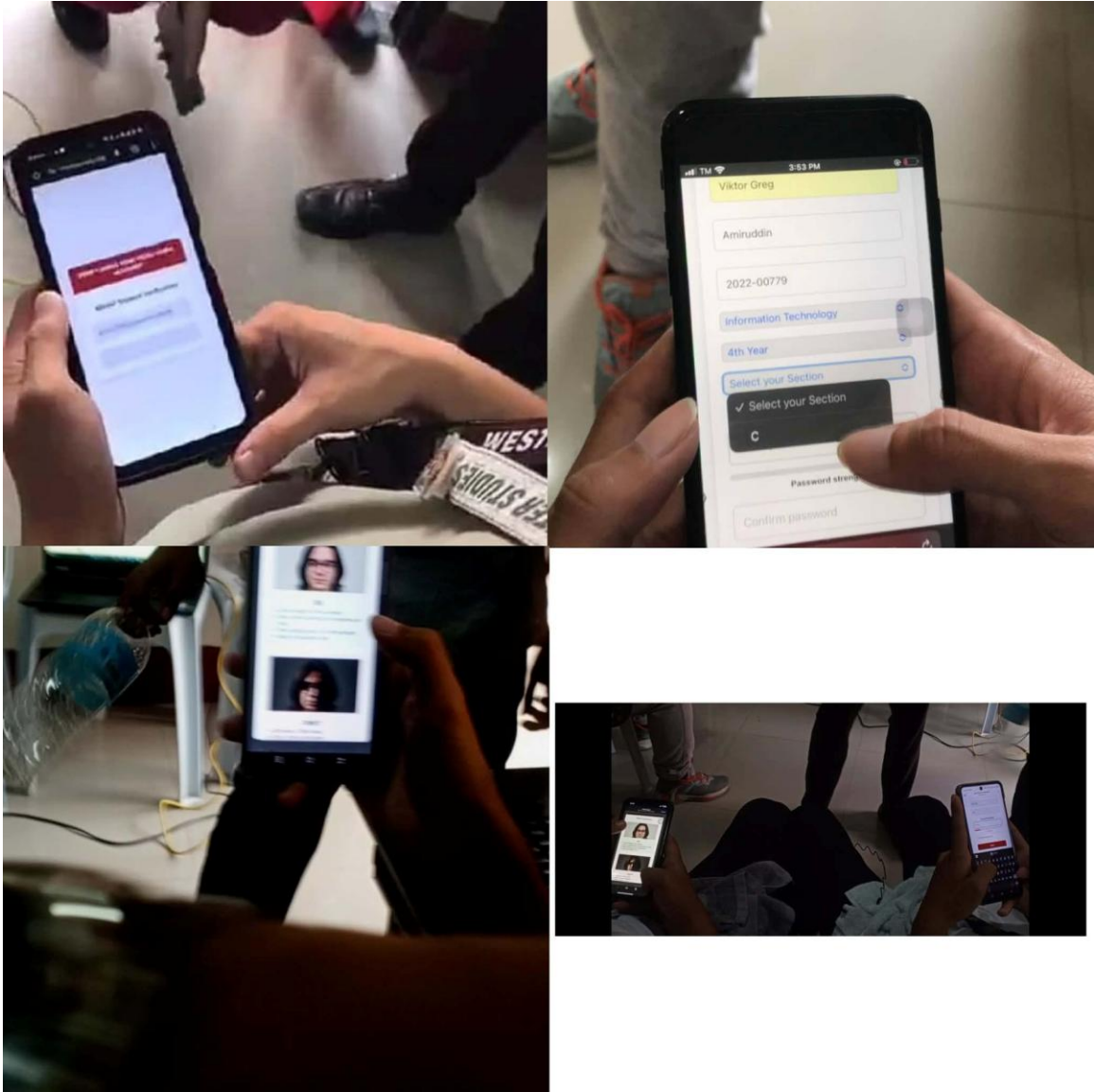
1479 Lab 1 is significantly larger than the specified system coverage. Because of this, a few
1480 areas in the room fell **outside the camera’s 102° horizontal Field of View (FOV)**.

1481 Despite this limitation, the system performed well within the visible zone of the camera.

1482 All students successfully registered their biometric data before the live test. The process
1483 included **face image uploading** and **biometric encoding**, both of which were completed
1484 without errors.

1485 **No bugs or errors were encountered** during the entire duration of the 2nd beta test. The
1486 system functioned smoothly and continuously, with the only minor delay caused by
1487 **internet speed fluctuations**, not by the system itself.

1488 **Testing Images**



1489

1490 Figure 21: Appendix A - Student Registering Biometrics (2nd Attempt)

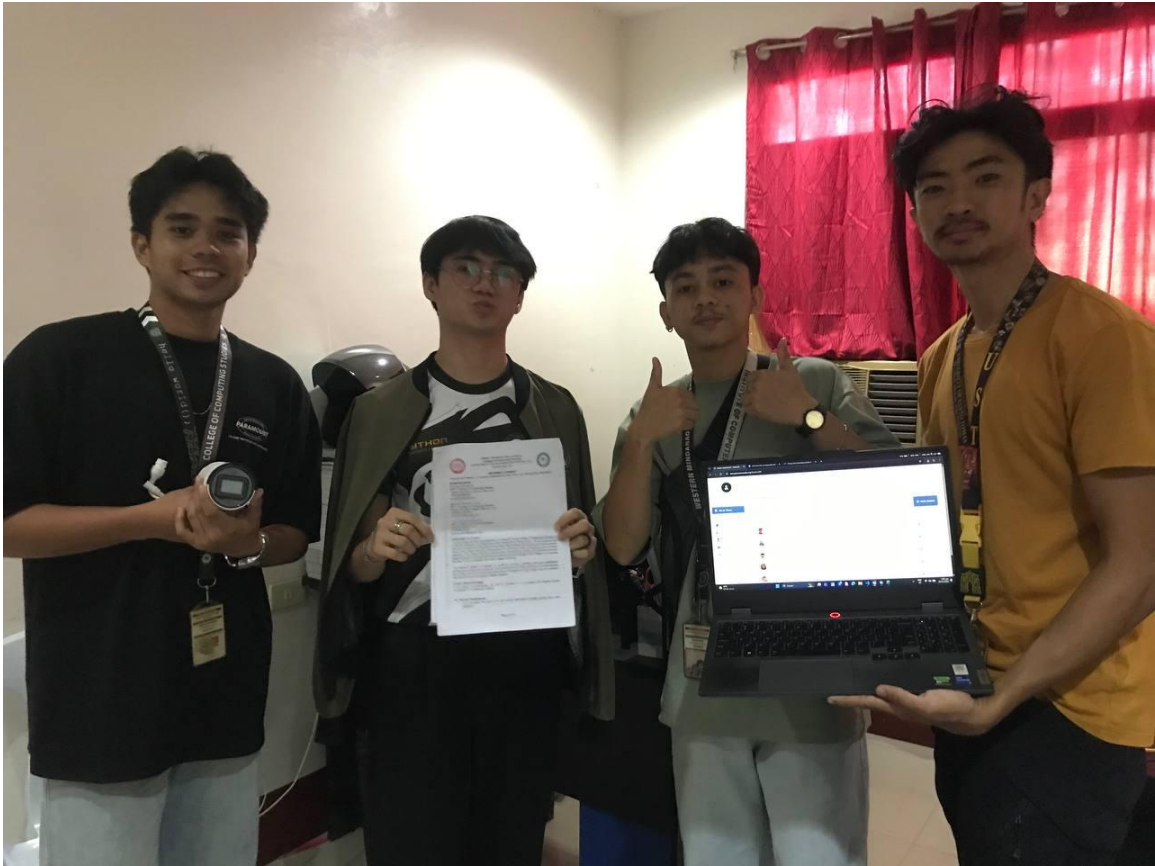


Figure 22: Appendix A - Group Photo with System Tester and Research Team (2nd Attempt)

Results of the 2nd Beta Testing

The results showed a significant improvement from the 1st attempt, with all functions working smoothly:

1. Student Biometric Registration

- All 15 students successfully completed registration.
- The **face encoding similarity** ranged from **87% to 95%**, which is **above the required 85% threshold**.
- All face embeddings were saved correctly in the database.

2. Recognition Performance During Attendance

- Fully registered students were **consistently recognized** within **1–3 meters**, which matches the system's optimal range.

1505 • Recognition occurred within **less than 1 second** after the student entered the
1506 FOV.

1507 **3. Bounding Box Stability**

1508 *Improvements from the 1st attempt were confirmed:*

- 1509 • **No more box jumping**
- 1510 • **No more box swapping**
- 1511 • Tracking remained stable even with multiple students simultaneously detected
- 1512 • As long as the student's face was not covered, the bounding box remained
- 1513 anchored to the correct identity

1514 **4. Liveness Attendance Flow**

- 1515 • Each student successfully triggered the liveness attendance capture, marking
- 1516 them as *Present* in real time.
- 1517 • No false liveness failures or false acceptances were detected.
- 1518 • The system automatically logged each student's attendance in the database and
- 1519 CSV backup.

1520 **User Evaluation Summary on 2nd Beta Testing (Likert Scale Results)**

1521 A Likert-scale evaluation was conducted among testers and instructors to assess the
 1522 Functionality, Usability, and Reliability of the system. Fifteen (15) items were rated using
 1523 a 5-point scale (1 = Strongly Disagree, 5 = Strongly Agree). The summarized results are
 1524 shown below.

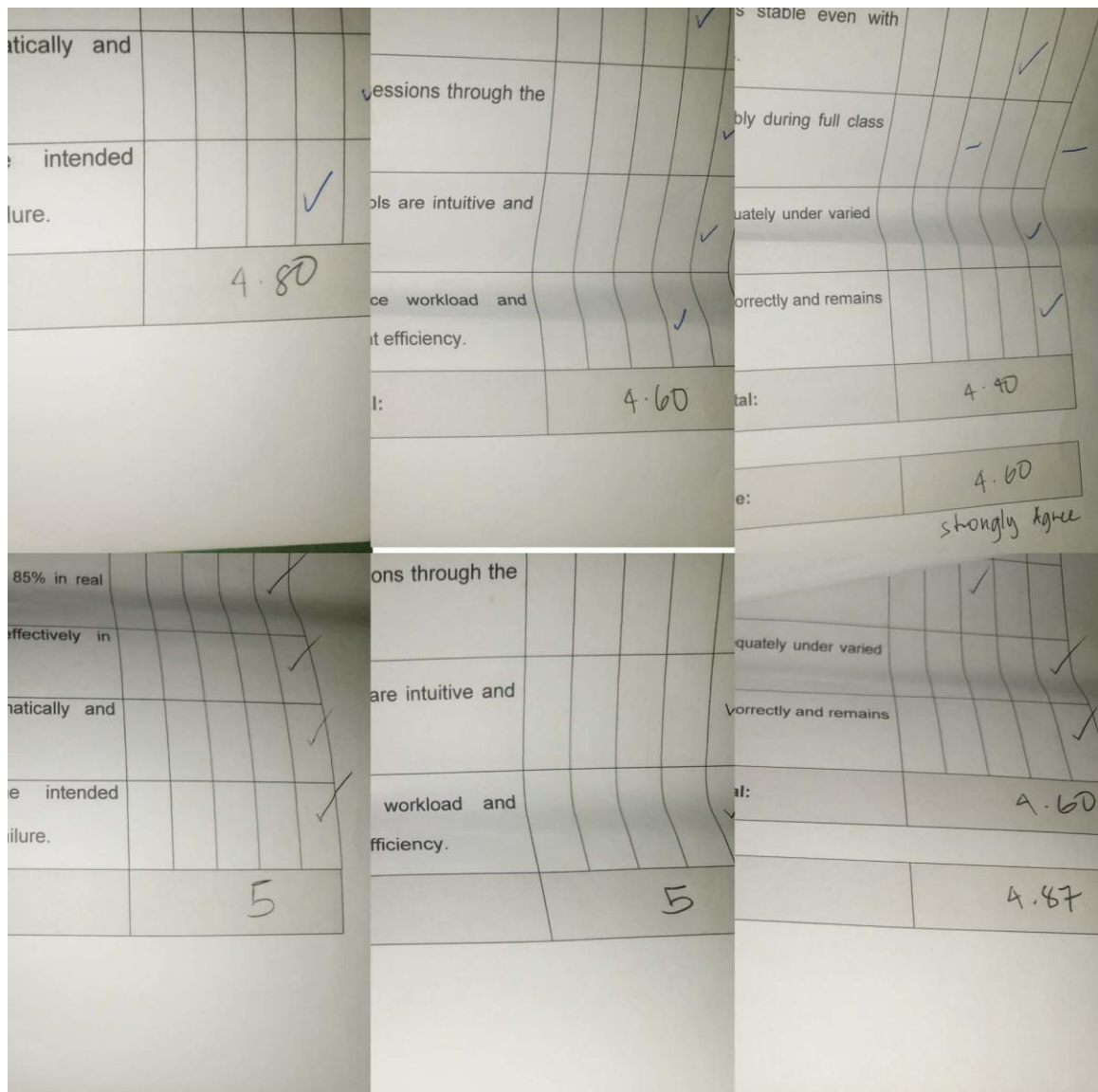


Figure 23: Appendix A – Beta Testing User Evaluation (2nd Attempt)

Category	Score
Functionality	4.80
Usability	4.60
Reliability	4.40
Overall Total:	4.60

1527 Table 17: Appendix A - Faculty Member Rating (2nd attempt)

Category	Score
Functionality	5.00
Usability	5.00
Reliability	4.60
Overall Total:	4.87

1528 Table 18: Appendix A - Student Rating (2nd Attempt)

1529 **Interpretation:**

1530 Both faculty and students rated the system as *highly functional*, *easy to use*, and *reliable*,
1531 confirming a significant improvement from the first beta test. The increased stability and
1532 accuracy of the system contributed to exceptional user satisfaction.

1533 **Summary of Findings (2nd Attempt)**

- 1534 • The system ran **error-free**, with zero bugs encountered.
- 1535 • Registration and recognition worked **smoothly and accurately**.
- 1536 • Bounding boxes remained **stable and precise** during the entire test.
- 1537 • Liveness detection was consistently successful for all students.
- 1538 • Recognition accuracy remained **above 87%**, validating the improved registration
1539 process.
- 1540 • User feedback showed **very high satisfaction**, indicating readiness for wider
1541 deployment.

1542 Informed consent participant's signature

...with the researcher. If you decide to take part in this study, you will be asked to sign a consent form. After you sign the consent form, you are still free to withdraw at any time and without giving a reason. Withdrawing from this study will not affect the relationship you have, if any, with the researcher. If you withdraw from the study before data collection is completed, your data will be returned to you or destroyed.

CONSENT

I have read and I understand the provided information and have had the opportunity to ask questions. I understand that my participation is voluntary and that I am free to withdraw at any time, without giving a reason and without cost. I understand that I will be given a copy of this consent form. I voluntarily agree to take part in this study.

Participant's signature VINCENZO A. TALANI Date 12/09

Researcher's signature Guilmar A. Quimba Date 12/09

1543

1544 Figure 24: Faculty Member Signature (2nd Attempt)

98 In this study. If you decide to take part in this study, you will be asked to sign a consent
99 form. After you sign the consent form, you are still free to withdraw at any time and without
100 giving a reason. Withdrawing from this study will not affect the relationship you have, if
101 any, with the researcher. If you withdraw from the study before data collection is
102 completed, your data will be returned to you or destroyed.

103
104
105
CONSENT

106 I have read and I understand the provided information and have had the opportunity
107 ask questions. I understand that my participation is voluntary and that I am free
108 withdraw at any time, without giving a reason and without cost. I understand that I will
109 be given a copy of this consent form. I voluntarily agree to take part in this study.

110
111 Participant's signature [Signature] Date 12/06/25

112
113 Researcher's signature Guilmar Date 12/06/25

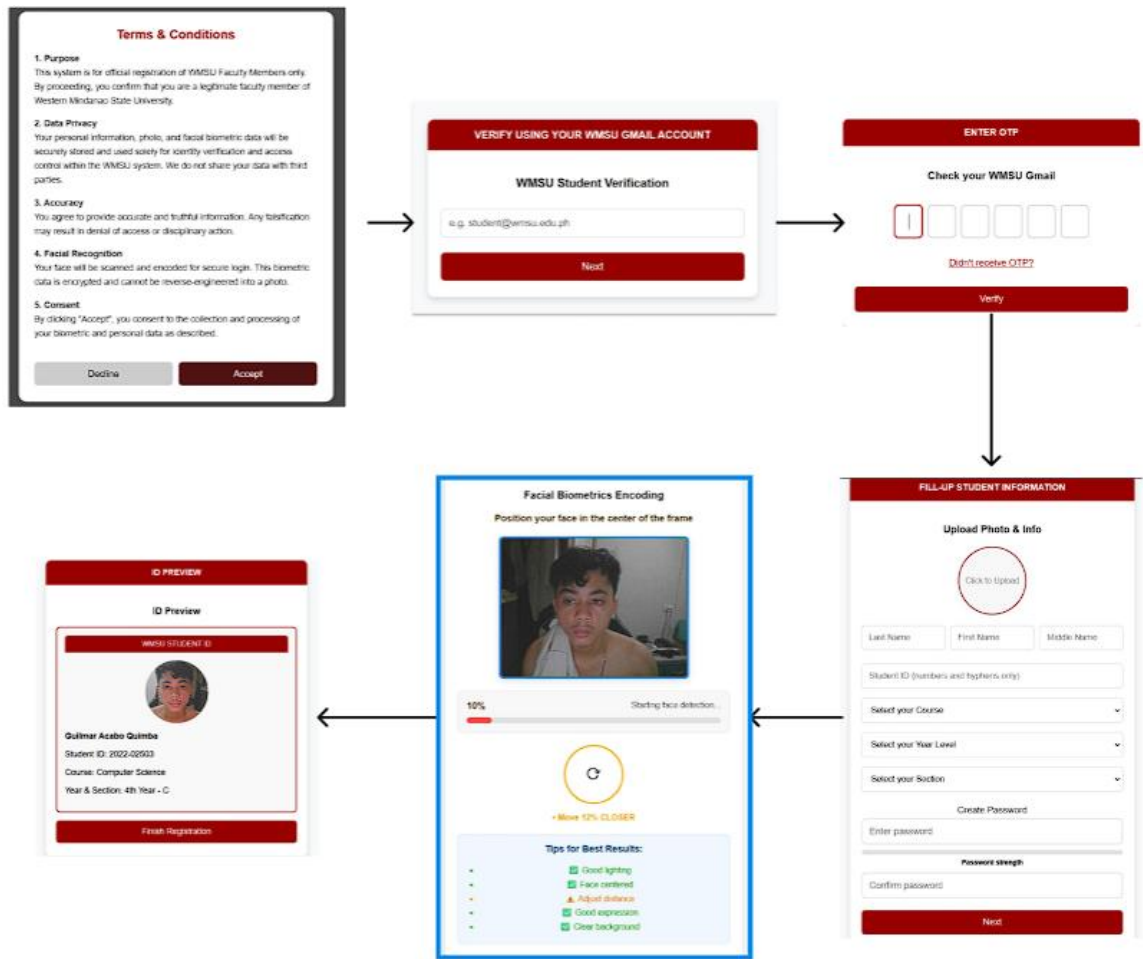
114
115
116

1545

1546 Figure 25: Student/s Signature (2nd Attempt)

1547

Appendix B: System Design



1548

1549 Figure 26: Appendix B - Registration Process Interface

1550 Description:

1551 This figure shows the complete registration workflow for new users.

- 1552 1. **Terms and Conditions Page** – Displays the system’s usage policy that the user
- 1553 must accept before proceeding.
- 1554 2. **WMSU Gmail Verification** – Requires the student to enter their official
- 1555 institutional email (studentid@wmsu.edu.ph).

- 1556 3. **OTP Verification Page** – Prompts the user to input the 6-digit one-time
1557 password sent to their email.
- 1558 4. **Fill-Up Information Form** – Collects the student’s personal and academic
1559 details.
- 1560 5. **Biometric Capture (Face Encoding)** – Captures the student’s face data for
1561 liveness detection and recognition.
- 1562 6. **Registration Summary Page** – Presents all submitted information and confirms
1563 readiness for the face attendance system.

The figure displays three mobile application screens for user authentication and account recovery. The first screen, titled 'WMSU Login', features a red header, a 'WMSU Email' input field with the placeholder 'yourname@wmsu.edu.ph', a 'Password' input field with the placeholder 'Enter your password', a red 'Login' button, and a red link for 'Forgot Password?'. The second screen, titled 'Reset Password', has a red header, a 'WMSU Email' input field with the placeholder 'eh202202503@wmsu.edu.ph', a red button labeled 'Send OTP (6s)', a red link for 'Back to Login', and a message: 'Password reset OTP sent to your email. Please wait 6 seconds before requesting another OTP.' The third screen, titled 'Verify OTP', has a red header, a message 'Enter the 6-digit code sent to your email:', an 'OTP Code' input field with the placeholder '123456', a 'New Password' input field with the placeholder 'Enter new password', a 'Confirm Password' input field with the placeholder 'Confirm new password', a red 'Reset Password' button, and a red link for 'Back'.

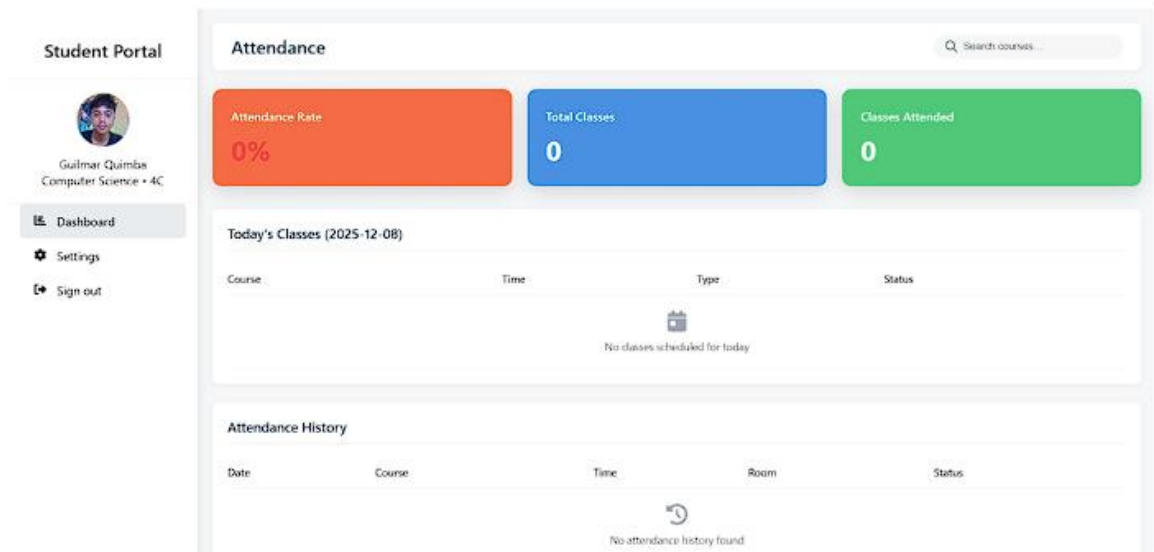
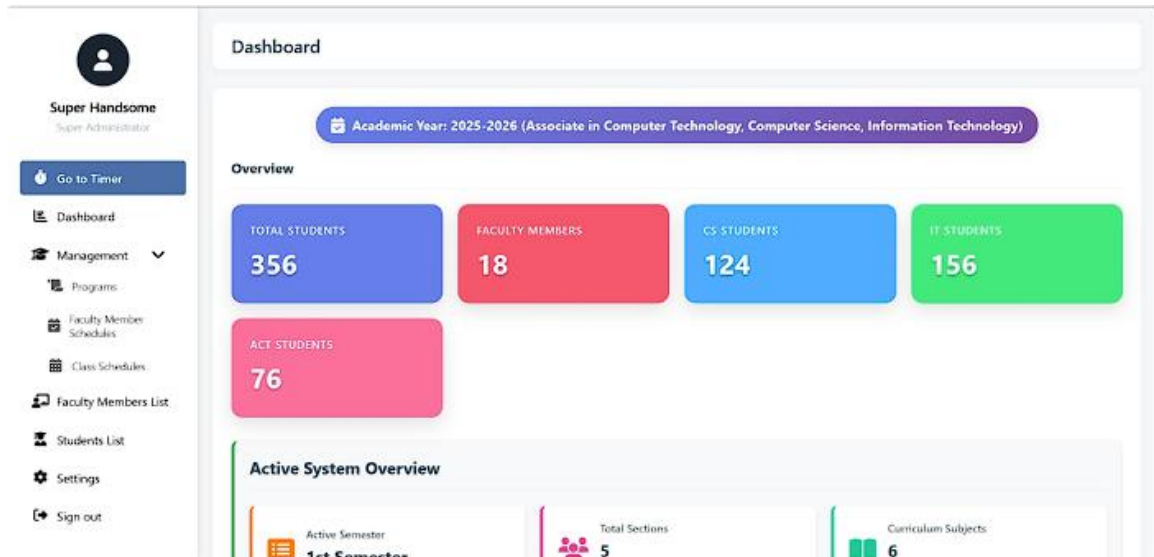
1564

1565 Figure 27: Appendix B - Login and Account Recovery Interfaces

1566 **Description:**

1567 This figure presents the login and password recovery process.

- 1568 1. **Login Screen** – Allows users to enter their email and password to access the
1569 system.
- 1570 2. **Reset Password UI** – Enables users to request a password reset via email.
- 1571 3. **OTP Verification & New Password Form** – Validates the OTP and allows users
1572 to set a new password.



1573

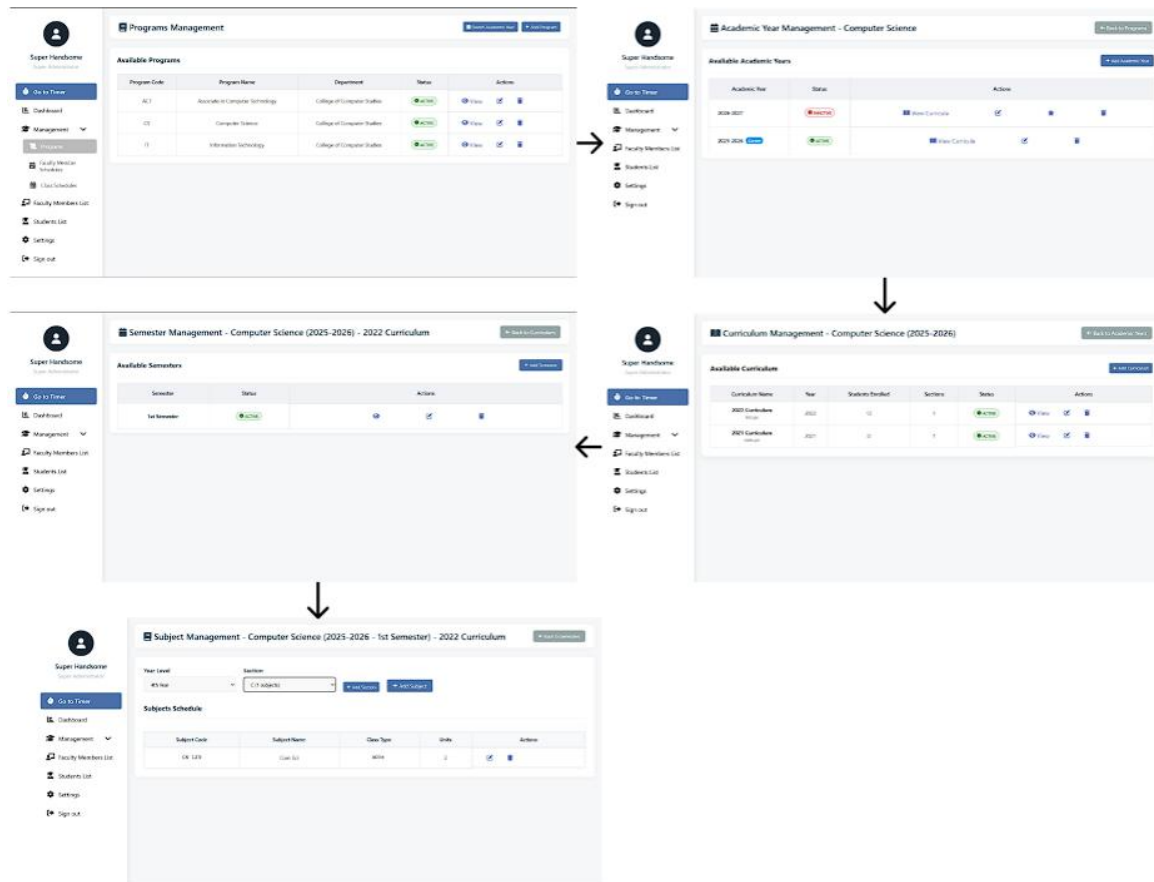
1574 Figure 28: Appendix B - Admin and Student Dashboard Interfaces

1575 **Description:**

1. **Admin Dashboard** – Displays total counts of Students, Faculty Members, and program-specific counts (CS, IT, ACT), including real-time system activity overview.

1578

1579 2. **Student Dashboard** – Shows the student’s attendance records and system
1580 notifications.



1581

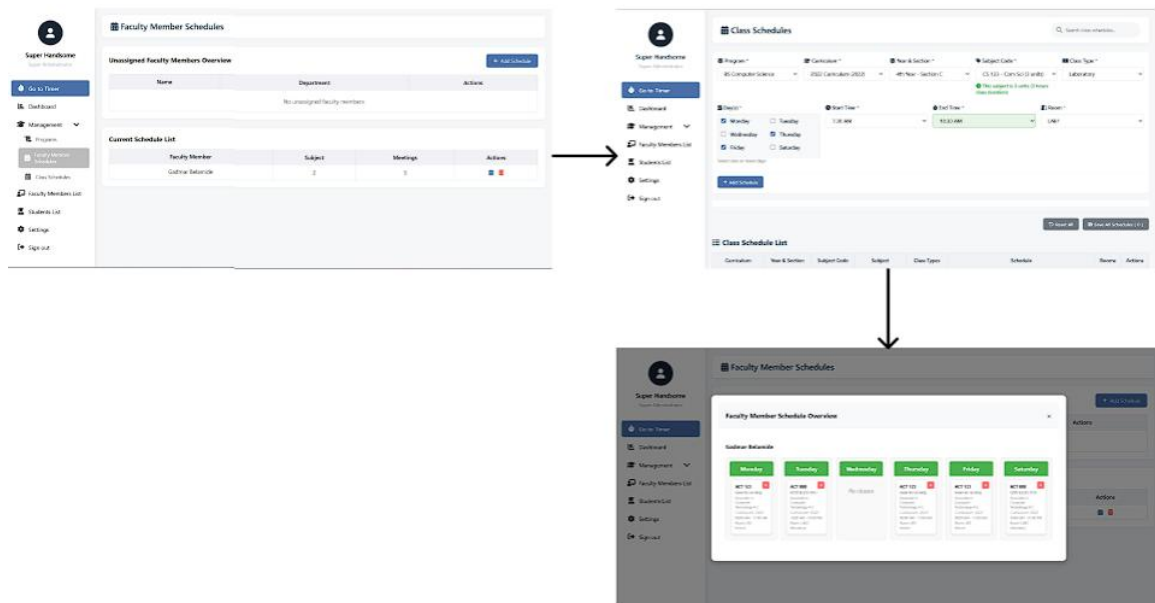
1582 Figure 29: Appendix B - Academic and Program Management Interfaces

1583 **Description:**

1584 This figure shows the admin interfaces used to configure academic structures.

- 1585 1. **Program Management Overview** – Allows management of academic programs
1586 offered by the institution.
- 1587 2. **Academic Year Management** – Used to organize and configure official academic
1588 years.
- 1589 3. **Curriculum Management** – Defines curriculum structures per program and year
1590 level.

- 1591 4. **Semester Management Interface** – Manages semester settings for each
 1592 academic year.
- 1593 5. **Subject Management Interface** – Handles subject creation, assignment, and
 1594 categorization.



1595

1596 Figure 30: Appendix B - Faculty and Class Scheduling Interfaces

1597 **Description:**

1598 This figure illustrates the management of faculty schedules and class timetable
 1599 configuration.

- 1600 1. **Faculty Member Schedule Overview** – Displays the teaching load and assigned
 1601 classes of faculty members.
- 1602 2. **Class Schedule Interface** – Used to add, edit, and manage class schedules.
- 1603 3. **Faculty Schedule Overview** – Provides a summarized view of faculty schedules.

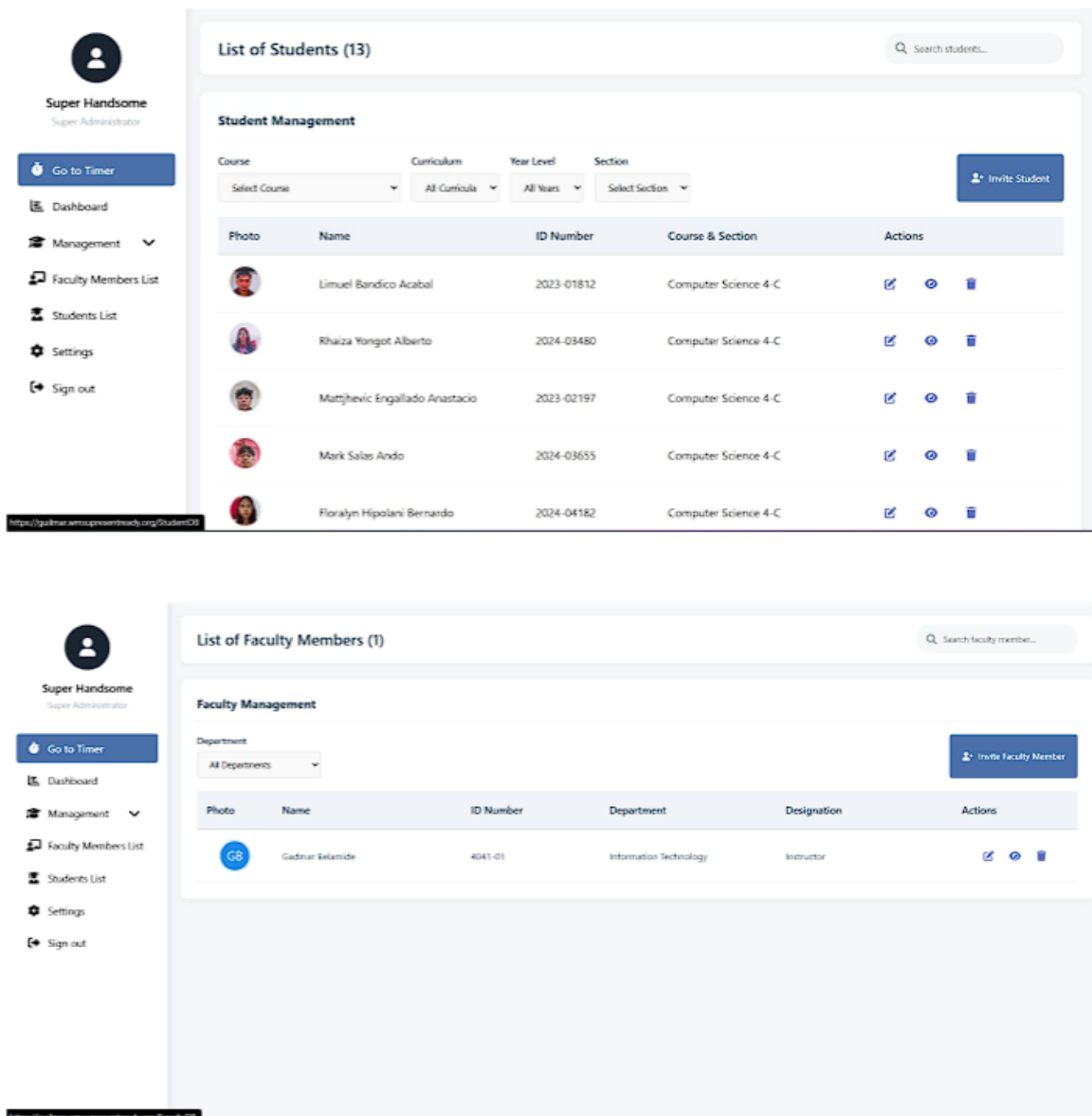


Figure 31: Appendix B - Faculty and Student List Interfaces

Description:

- Faculty List UI** – Displays all registered faculty members with options for CRUD operations.
- Student List UI** – Shows all registered students together with their enrollment details.

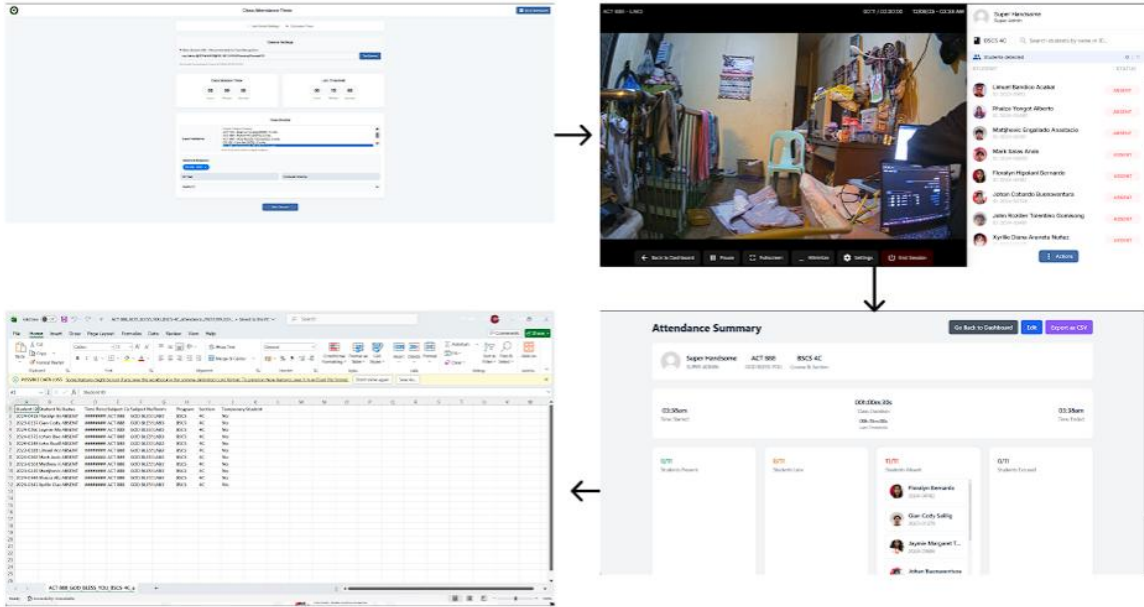


Figure 32: Appendix B - Go-To-Timer and Live Attendance Interfaces

Description:

This figure shows the complete workflow of initiating and conducting real-time attendance.

- Go-to Timer Setup UI** – Includes RTSP camera selection, class session timer, late threshold configuration, and class details.
- Live Class Session UI** – Displays the camera feed, face detection and recognition results, faculty name, student list, and real-time attendance marking.
- Session Summary Screen** – Shows the summary of attendance after the class session ends.
- CSV Export Output** – The generated .csv attendance file after clicking “Save”.

1622

Appendix C: Evaluation Tool

1623 Instructions:

1624 This questionnaire aims to gather your level of agreement regarding the Functionality,
 1625 Usability, and Reliability of the “**Present and Ready: A Liveness Detection for Real-**
 1626 **Time Face Recognition Attendance System.**”

1627 After using or testing the system, please evaluate each statement by selecting the
 1628 response that best reflects your experience.

1629 *Likert Scale:*

1630 **1 – Strongly Disagree | 2 – Disagree | 3 – Neutral | 4 – Agree | 5 – Strongly Agree**

1631 FUNCTIONALITY

Item No.	Statement	1	2	3	4	5
1	The system accurately detects faces on 1-3 meters in real time.					
2	The system correctly recognizes registered students with an accuracy of 85% in real time.					
3	Liveness detection works effectively in preventing spoof attempts.					
4	Attendance is logged automatically and reflects accurate timestamps.					
5	The system performs the intended attendance functions without failure.					
Functionality Scale Total:						

1632

Table 19: Evaluation Tool - Likert Scale (Functionality)

1633 **USABILITY**

Item No.	Statement	1	2	3	4	5
6	The system is easy to learn and operate.					
7	The user interface is structured, clear, and easy to navigate.					
8	Starting and ending class sessions through the system is simple.					
9	System features and controls are intuitive and understandable.					
10	The system helps reduce workload and improves class management efficiency.					
Usability Scale Total:						

1634 Table 20: Evaluation Tool - Likert Scale (Usability)

1635 **RELIABILITY**

Item No.	Statement	1	2	3	4	5
11	The system consistently records attendance without interruptions as long as there is a strong internet connection.					
12	Facial recognition remains stable even with natural student movements.					
13	The system operates reliably during full class sessions (10–15 students).					
14	The system performs adequately under varied classroom conditions.					
15	Attendance data is stored correctly and remains accessible at all times.					
Reliability Scale Total:						

1636 Table 21: Evaluation Tool - Likert Scale (Reliability)

Overall Total Scale:	
-----------------------------	--

1637 Table 22: Evaluation Tool - Likert Scale (Overall Total Scale)

1638

Appendix D: Relevant Source Code

```
# FACE DETECTION FOR SIDE/PROFILE FACES
faces = []
detection_method = "insightface"

try:
    # First try InsightFace with default settings
    faces = face_analysis.get(img, max_num=3)

    # If no faces found, try with lower detection threshold for profile faces
    if not faces and hasattr(face_analysis.det_model, 'det_thresh'):
        original_threshold = face_analysis.det_model.det_thresh
        try:
            # Lower threshold for better side face detection
            face_analysis.det_model.det_thresh = 0.2
            faces = face_analysis.get(img, max_num=3)
            face_analysis.det_model.det_thresh = original_threshold

            if faces:
                detection_method = "insightface_low_thresh"
                logger.debug(f"Face detected with low threshold ({len(faces)} faces)")
        except Exception as e:
            logger.warning(f"Failed to adjust InsightFace threshold: {e}")
            if hasattr(face_analysis.det_model, 'det_thresh'):
                face_analysis.det_model.det_thresh = original_threshold
```

1639

1640

Figure 33: Appendix D - InsightFace 512-D Embedding Generation

1641

- This illustrates how the system extracts a face crop from the video frame and processes it through the InsightFace encoder to generate a **512-dimensional facial embedding**, a numerical representation that uniquely describes the person in the image. This embedding becomes the foundation for all subsequent recognition steps because it converts visual facial features into measurable, machine-comparable data.

1645

1646

1647

```

# CHECK for better accuracy
if similarities.size > 1:
    sorted_indices = np.argsort(similarities[::-1])
    best_sim = similarities[sorted_indices[0]]
    second_best_sim = similarities[sorted_indices[1]]

    similarity_gap = best_sim - second_best_sim
    if similarity_gap < 0.12:
        if DEBUG_MODE:
            logger.debug(f" AMBIGUOUS: Best match gap too small ({similarity_gap:.3f})")
            name = "Unknown - Ambiguous match"
            continue

```

1648

1649 Figure 34: Appendix D - Cosine Similarity Computation

- 1650 • The Cosine Computation shows the similarity comparison between two facial
- 1651 embeddings using **cosine similarity**, where values closer to **1.0** indicate a strong
- 1652 identity match. By comparing the new embedding against stored profiles, the
- 1653 system determines how closely two faces resemble each other, allowing it to verify
- 1654 or reject an identity based on mathematical distance rather than raw pixel
- 1655 comparison.


```

# ADAPTIVE RECOGNITION THRESHOLDS based on orientation
if face_orientation == "front":
    # Frontal face - normal thresholds
    if pixel_distance_from_center > 400: # Very far
        recognition_threshold = 0.57
        min_face_size = 50
    elif pixel_distance_from_center > 300: # Far
        recognition_threshold = 0.59
        min_face_size = 45
    elif pixel_distance_from_center > 200: # Medium
        recognition_threshold = 0.61
        min_face_size = 40
    else: # Close
        recognition_threshold = 0.63
        min_face_size = 35
else:
    # Side/profile face - LOWER THRESHOLDS
    if pixel_distance_from_center > 400: # Very far
        recognition_threshold = 0.56 # Lower for side faces
        min_face_size = 50
    elif pixel_distance_from_center > 300: # Far
        recognition_threshold = 0.58 # Lower for side faces
        min_face_size = 45
    elif pixel_distance_from_center > 200: # Medium
        recognition_threshold = 0.60 # Lower for side faces
        min_face_size = 40
    else: # Close
        recognition_threshold = 0.62 # Lower for side faces
        min_face_size = 35

logger.debug(f" Side face detection - using lower threshold: {recognition_threshold:.3f}")

```

Figure 35: Appendix D - Recognition Threshold Tuning

- This demonstrates the process of setting and adjusting the **recognition threshold**, the minimum similarity score required to consider two faces a match. A properly tuned threshold reduces both false positives and false negatives, ensuring that the recognition module performs reliably within the detection pipeline and produces accurate identity decisions during live video testing.

Appendix E: Ethical Clearance

WMSU-REO-CE-0012
Effective Date: 20-June-2025

WESTERN MINDANAO STATE UNIVERSITY
Research and Extension
Research Ethics Office
accredited by
PHILIPPINE HEALTH RESEARCH ETHICS BOARD

grants
this

CERTIFICATE OF EXEMPTION

to

Rhodmin Lou B. Berioso
Guilmar A. Quimba
Lawrence Olivier N. Tilde

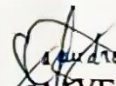
Research Title: **PRESENT AND READY: A LIVENESS DETECTION FOR
REAL TIME FACE RECOGNITION ATTENDANCE**

WMSU REO Code: **2026-UG-0149**

This certifies that this protocol does not involve human participants or with no or minimal risk to human participants. Hence, this is an exemption for research ethics review. The study may proceed with implementation.

Compliance on the standard conditions attached here is required. Any change in the protocol invalidates the certificate and will require a new application for review.

Issued this 10th day of February 2026, at Research Ethics Office, Western Mindanao State University, Zamboanga City, Philippines.


ANALYN D. SAAVEDRA, Ph.D., J.D.
Research Ethics Director

1665 **Appendix H: Curriculum Vitae**

1666 **Personal Background**

1667 Full Name: Rhodmin Lou B. Berioso

1668 Place of Birth: Zamboanga City

1669 Contact Number: 09263126144

1670 Email Address: rhodmin13@gmail.com

1671 Date of Birth: March 5, 2004

1672 Age: 21

1673 Gender: Male

1674 Citizenship: Filipino

1675 Status: Single



1676 **Educational Background**

Degree Earned	School/Degree	Year Graduated
Elementary	Southern Support Command Elementary School	2016
High School	Zamboanga National High School West	2020
Senior High School	Baliwasan Senior High School	2022

1677

1678 **Awards/Honors/Licensed Received**

1679 2020-2022 With Honors (Senior High School)

1680 2010-2016 Achiever (Elementary)

1681 **Personal Background**

1682 Full Name: Guilmar A. Quimba

1683 Place of Birth: Zamboanga City

1684 Contact Number: 09364885993

1685 Email Address: qguilmar@gmail.com

1686 Date of Birth: November 23, 2003

1687 Age: 21

1688 Gender: Male

1689 Citizenship: Filipino

1690 Status: Single



1691 **Educational Background**

Degree Earned	School/Degree	Year Graduated
Elementary	Canelar Elementary School	2016
High School	Zamboanga National High School West	2020
Senior High School	Baliwasan Senior High School	2022

1692

1693 **Special Skills and Hobbies:**

1694 - Guitarist

1695 - Gamer

1696 - Programming

1697 **Personal Background**

1698 Full Name: Lawrence Olivier N. Tilde

1699 Place of Birth: Zamboanga City

1700 Contact Number: 09555412051

1701 Email Address: lawrencetilde@gmail.com

1702 Date of Birth: October 08, 2003

1703 Age: 22

1704 Gender: Male

1705 Citizenship: Filipino

1706 Status: Single



1707 **Educational Background**

Degree Earned	School/Degree	Year Graduated
Elementary	Zamboanga Puericulture Center School	2016
High School	Science, Technology, Engineering and Mathematics	2020
Senior High School	Science, Technology, Engineering and Mathematics	2022

1708

1709 **Special skills and Hobbies:**

1710 - Hiking

1711 - Cycling

1712 - Swimming

1713 - Programming